SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS OF THE DEGREE OF

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

BY

Licia Almeida
Janaki Bal
Shravani Jadhav
Priyadarshini Sandilyan

UNDER THE GUIDANCE OF

Prof. Jaya Jeswani

(Department of Information Technology)

INFORMATION TECHNOLOGY DEPARTMENT

XAVIER INSTITUTE OF ENGINEERING

UNIVERSITY OF MUMBAI

2025 - 2026

XAVIER INSTITUTE OF ENGINEERING
MAHIM CAUSEWAY, MAHIM,MUMBAI - 400016.

CERTIFICATE

This to certify that,

Licia Almeida	202203005
Janaki Bal	202203007
Shravani Jadhav	202203028
Priyadarshini Sandilyan	202203004

Have satisfactorily carried out the MAJOR-PROJECT work titled ***Heat Stress*** in partial fulfillment of the degree of Bachelor of Engineering as laid down by the University of Mumbai during the academic year 2025-2026.

Prof. Jaya Jeswani
Supervisor / Guide

Dr.Puja Padiya
Head of Department

Dr. Lata Ragha
Principal

PROJECT REPORT APPROVAL FOR B.E.

This project entitled Heat Stress
By

Name	Roll No.
Licia Almeida	(202203005)
Janaki Bal	(202203007)
Shravani Jadhav	(202203028)
Priyadarshini Sandilyan	(202203004)

is approved for the degree of BACHELOR OF ENGINEERING.

Examiners

1.

2.

Supervisors

1.

2.

Date:
Place: MAHIM, MUMBAI

Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources.

We also declare that I have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission.

We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which thus have not been properly cited or from whom proper permission have not been taken when needed.

Name	**Signature**
Licia Almeida (202203005)	
Janaki Bal (202203007)	
Shravani Jadhav (202203028)	
Priyadarshini Sandilyan (202203004)	

Date:

Abstract

Heat stress is a growing environmental and public health challenge, especially in densely populated and rapidly urbanizing cities like Mumbai. The combined effects of rising temperatures, high humidity, and urban heat island phenomena have significantly increased the intensity and frequency of extreme heat events. These conditions not only affect human comfort but also pose serious risks to health, productivity, and overall urban sustainability. Traditional temperature-based analysis is insufficient to fully capture human heat stress, as it does not account for critical atmospheric factors such as humidity, solar radiation, and wind speed. Therefore, advanced heat stress indices such as the Heat Index (HI) and Wet-Bulb Globe Temperature (WBGT) are required to provide a more accurate representation of how heat is experienced by the human body. This project presents a comprehensive and integrated framework for heat stress analysis and prediction in Mumbai using two complementary datasets. The first part of the study is based on long-term historical data obtained from the India Meteorological Department (IMD), covering the period from 1969 to 2025 for two key stations: Santacruz and Colaba. This dataset is used to analyze historical trends, compute HI More advanced indices such as Heat Index (HI). The second part of the project utilizes ERA5 reanalysis data spanning from 2015 to 2026 to develop a predictive modeling framework for heat stress. In this implementation, WBGT is used as the primary target variable, as it is an internationally recognized standard for assessing physiological heat stress. Advanced machine learning and deep learning models are applied to predict WBGT using a wide range of meteorological variables and engineered features. The project integrates data preprocessing, feature engineering, imputation techniques, class balancing, and temporal validation strategies to ensure robust and reliable results. It also incorporates explainability and causal analysis techniques to identify key drivers of heat stress and improve model interpretability. By combining long-term climatic analysis with modern predictive modeling approaches, this study provides valuable insights into heat stress patterns in Mumbai and offers a scalable framework that can support early warning systems, public health planning, and climate adaptation strategies.

Acknowledgement

We would like to thank Fr. Dr. John Rose S.J (Director of XIE) for providing us with such an environment so as to achieve goals of our project and supporting us constantly.

We express our sincere gratitude to our Honorable Principal Dr. Lata Ragha for encouragement and facilities provided to us.

We would like to place on record our deep sense of gratitude to Dr. Puja Padiya Head of Deptartment Of Information Technology, Xavier Institute of Engineering, Mahim, Mumbai, for his generous guidance help and useful suggestions.

With deep sense of gratitude we acknowledge the guidance of our project guide Guide Name. The time-to-time assistance and encouragement by her has played an important role in the development of our project.

We would also like to thank our entire Information Technology staff who have willingly cooperated with us in resolving our queries and providing us all the required facilities on time.

Licia Almeida (202203005)	____________________
Janaki Bal (202203007)	____________________
Shravani Jadhav (202203028)	____________________
Priyadarshini Sandilyan (202203004)	____________________

List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
WBGT	Wet Bulb Globe Temperature
HI	Heat Index
RH	Relative Humidity
DBT	Dry Bulb Temperature
WBT	Wet Bulb Temperature
DPT	Dew Point Temperature
AT	Apparent Temperature
SLP	Station Level Pressure
MSLP	Mean Sea Level Pressure
WS	Wind Speed
WD	Wind Direction
ERA5	Fifth Generation ECMWF Reanalysis Dataset
IMD	India Meteorological Department
EDA	Exploratory Data Analysis
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
LSTM	Long Short-Term Memory
XGBoost	Extreme Gradient Boosting
KNN	K-Nearest Neighbors
SVC	Support Vector Classifier
LDA	Linear Discriminant Analysis
BBAG	Balanced Bagging
SAINT	Self-Attention and Intersample Transformer
PI	Prediction Interval
UHI	Urban Heat Island
UTC	Coordinated Universal Time
IST	Indian Standard Time

List of Figures

List of Tables

Contents

Chapter 1
Introduction

1.1 Introduction

Heat stress is a growing environmental and public health challenge, especially in densely populated and rapidly urbanizing cities like Mumbai. The combined effects of rising temperatures, high humidity, and urban heat island phenomena have significantly increased the intensity and frequency of extreme heat events. These conditions not only affect human comfort but also pose serious risks to health, productivity, and overall urban sustainability. Traditional temperature-based analysis is insufficient to fully capture human heat stress, as it does not account for critical atmospheric factors such as humidity, solar radiation, and wind speed. Therefore, advanced heat stress indices such as the Heat Index (HI) and Wet-Bulb Globe Temperature (WBGT) are required to provide a more accurate representation of how heat is experienced by the human body. The second part of the project utilizes ERA5 reanalysis data spanning from 2015 to 2026 to develop a predictive modeling framework for heat stress. In this implementation, WBGT is used as the primary target variable, as it is an internationally recognized standard for assessing physiological heat stress. Advanced machine learning and deep learning models are applied to predict WBGT using a wide range of meteorological variables and engineered features. The project integrates data preprocessing, feature engineering, imputation techniques, class balancing, and temporal validation strategies to ensure robust and reliable results. It also incorporates explainability and causal analysis techniques to identify key drivers of heat stress and improve model interpretability. By combining long-term climatic analysis with modern predictive modeling approaches, this study provides valuable insights into heat stress patterns in Mumbai and offers a scalable framework that can support early warning systems, public health planning, and climate adaptation strategies.

1.2 Problem Definition

Rapid urbanization, population growth, and global climate change have significantly intensified the occurrence of extreme heat events in Mumbai. The city's dense infrastructure, reduced green cover, and coastal humidity amplify heat retention, leading to prolonged periods of thermal discomfort and increased health risks such as heat exhaustion, heatstroke, and cardiovascular stress. Conventional temperature-based assessments are inadequate for evaluating real-world heat stress, as they fail to incorporate the combined influence of key atmospheric parameters such as relative humidity, solar radiation, and wind speed. As a result, such approaches provide an incomplete and sometimes misleading representation of human thermal stress. Furthermore, analyzing long-term climatic data presents additional challenges, including missing values, inconsistencies across datasets, and the presence of rare extreme heat events that lead to class imbalance in modeling tasks. In predictive frameworks, the complexity of atmospheric interactions and temporal dependencies further complicates accurate forecasting. Therefore, there is a critical need for a comprehensive and robust framework that can effectively analyze long-term heat stress trends using reliable historical data, accurately quantify human heat stress using scientifically validated indices, and predict future heat stress conditions using advanced machine learning techniques. Such a framework must also address data-related challenges such as missing values, imbalance, and temporal dependencies to ensure reliable and realistic modeling outcomes. Addressing these challenges is essential for developing effective heat stress assessment systems that can support public health preparedness and climate resilience.

1.3 Scope of the Project

This project presents a comprehensive framework for analyzing and predicting heat stress conditions in urban environments. It integrates both historical meteorological data and advanced reanalysis datasets to capture long-term trends as well as future patterns. The approach combines domain-specific heat stress indices with modern machine learning and deep learning techniques for accurate and reliable modeling. The overall methodology is designed to be scalable, interpretable, and applicable to real-world climate and public health challenges.

- Covers both historical analysis (IMD) and predictive modeling (ERA5).
- Focuses on Mumbai region using station-based and reanalysis datasets.
- Includes heat stress indices such as HI computation of Heat Index (HI) and WBGT.
- Applies machine learning and deep learning models for prediction.
- Incorporates feature engineering, class balancing, and temporal validation.
- Supports applications in urban planning, public health, and climate adaptation.
- Can be extended to other cities and regions with similar climatic conditions

1.4 Objectives

This project aims to develop a robust and data-driven framework for understanding and predicting heat stress conditions in Mumbai. It focuses on combining traditional meteorological analysis with advanced machine learning techniques to improve accuracy and reliability. The objectives are designed to address both analytical and predictive aspects of heat stress modeling. The overall goal is to create an interpretable and practical system that can support climate-related decision-making.

- To analyze long-term heat stress trends in Mumbai using IMD historical data (1969–2025).
- To compute and classify heat stress using HI.
- To develop an accurate prediction model for WBGT using ERA5 reanalysis data.
- To compare multiple machine learning and deep learning models for heat stress prediction.
- To perform feature engineering and selection to identify key drivers of heat stress.
- To handle data challenges such as missing values, imbalance, and temporal dependencies.
- To provide an interpretable and reliable framework for heat stress assessment.

Chapter 2
Literature Survey

2.1 Survey of Existing Systems

The existing body of research on heat stress and thermal regulation primarily utilizes machine learning and statistical modeling to predict thermal indices and environmental variables. The systems can be categorized into three main approaches:

- **Machine Learning Based Models:** Several studies employ algorithms like CatBoost, Random Forest (RF), and XGBoost to predict indices such as the Universal Thermal Climate Index (UTCI) and Predicted Mean Vote (PMV). For instance, a 2025 study in Tabriz utilized these models combined with Bayesian optimization and SHAP explainability to achieve accuracies between 0.72 and 0.83.
- **Deep Learning & Time Series Forecasting:** Advanced architectures like iTransformer (Inverted Transformers) have been proposed to handle multivariate time-series forecasting. By treating variables as tokens rather than time steps, these models capture multivariate correlations more effectively than traditional models like Autoformer or DLinear.
- **Hybrid Signal Decomposition:** Some systems use signal decomposition techniques such as Empirical Mode Decomposition (EMD) and Ensemble EMD (EEMD) followed by Multi-Layer Perceptrons (MLP) or Support Vector Regression (SVR) to forecast long-term solar radiation in Indian cities, achieving high R^2 values around 0.99.
- **Global Climate Model (GCM) Projections:** Research has also been conducted using CMIP5 global climate models to project future heat stress indices (HI) and their impact on human work productivity across India in response to global warming.

2.2 Limitations of Existing Systems & Research Gaps

While existing systems show high performance in specific contexts, they face several critical limitations:

- **Temporal and Seasonal Constraints:** Many existing frameworks focus exclusively on specific periods, such as the summer season, which limits their applicability for year-round heat stress management.

- **Geographical Specificity:** Most models are developed for specific urban canyons or regional datasets (e.g., specific cities in Iran or Pakistan), making their results difficult to adapt to other climatic regions without extensive retraining.
- **Data Dependency:** High-performing models like iTransformer are heavily dependent on the availability of large, high-dimensional multivariate datasets. Their training complexity increases significantly with very high-dimensional data.
- **Static Historical Analysis:** Several Indian studies rely solely on historical data (e.g., 2000–2012), leaving their generalizability over rapidly changing future climate trends untested.
- **Computational Overhead:** Complex ensembles and deep learning models require substantial computational resources for optimization and training, which can be a barrier for real-time local deployment.

2.2.1 Research Gaps

The analysis of current literature identifies the following research gaps, which your project aims to address:

- **Lack of Localized High-Resolution Models for Mumbai:** While broad studies exist for India, there is a gap in high-resolution, city-specific heat stress forecasting that combines both reanalysis data (like ERA5) and local observational data specifically for the Mumbai metropolitan region.
- **Integration of Diverse Heat Indices:** Many studies focus on a single index (like UTCI or HI). There is a need for systems that concurrently compute and forecast multiple derived indices—such as WBGT (Wet Bulb Globe Temperature), Humidex, and Apparent Temperature—to provide a comprehensive thermal stress profile.
- **Explainability in Urban Heat Forecasting:** There is limited research on providing local stakeholders with explainable ML results that identify which specific climatic variables (e.g., soil moisture, boundary layer height, or solar radiation) are driving extreme heat stress events in a coastal tropical city.
- **Real-time Adaptive Forecasting:** Most existing systems are diagnostic rather than prognostic. A gap exists for an end-to-end pipeline that installs dependencies, merges live observational data with historical reanalysis, and provides automated classification of heat stress days (Normal, Caution, Danger, Extreme).

Chapter 3
Data Collection & Preprocessing

3.1 Data Sources

3.1.1 IMD Dataset

The data used in this study was sourced from the India Meteorological Department (IMD), the principal government authority responsible for weather monitoring, forecasting, and climatological data management across India. IMD maintains a dense network of observation stations equipped with standardized instruments that record key meteorological parameters such as temperature, humidity, wind speed, rainfall, and solar radiation at regular intervals. The reliability, consistency, and long-term continuity of IMD datasets make them highly suitable for scientific research, particularly for climate variability and heat stress analysis. To capture the spatial variability of Mumbai's climate, data from two strategically located IMD weather stations were utilized: Colaba Weather Station and Santacruz Weather Station.

The Colaba station is situated in the southern part of Mumbai, close to the Arabian Sea. Due to its coastal location, it is strongly influenced by maritime conditions. This results in Moderated temperatures with lower diurnal variation, Higher relative humidity levels, Strong sea-breeze effects that influence local weather patterns These characteristics make Colaba ideal for representing coastal climatic conditions, especially important in understanding how proximity to large water bodies affects heat stress levels.

The Santacruz station is located in the suburban inland region of Mumbai, near the airport. Unlike Colaba, it is less influenced by direct coastal moderation and instead reflects urban land-based conditions. Key characteristics include:

- Higher daytime temperatures due to urban heat island effects
- Greater temperature variability
- Lower humidity compared to coastal regions at certain times

This station effectively captures inland urban climatic behavior, making it crucial for analyzing heat stress in densely built-up environments.

ERA5 Dataset

The heat stress project in Mumbai utilizes a comprehensive set of parameters derived from ERA5 reanalysis data, along with calculated thermal indices and engineered features to enhance model accuracy and predictive performance. The core ERA5 parameters include key meteorological variables such as 2-meter air temperature (t2m), also known as dry bulb temperature, 2-meter dew point temperature (d2m), 10-meter wind components (u10 and v10) used to calculate total wind speed, surface pressure (sp), mean sea-level pressure (msl), total precipitation (tp), and surface solar radiation downwards (ssrd).

Using these primary variables, several important thermal indices are calculated to better represent heat stress conditions. These include the Wet Bulb Globe Temperature (WBGT), which serves as the main target variable for heat stress modeling, the Heat Index (HI), which is computed using temperature and relative humidity, and Relative Humidity (RH), derived from t2m and d2m values.

In addition to raw and derived meteorological data, the project incorporates a range of engineered features to improve the robustness of the model. Temporal features such as year, month, day, hour, and day of the week are included to capture time-based patterns. Cyclical time encoding is applied using sine and cosine transformations of hour and month to effectively model daily and seasonal variations. Lagged features, such as previous values of WBGT at time intervals like t-1, t-3, and t-6 hours, are used to capture temporal dependencies. Rolling statistics, including moving averages and standard deviations (for example, a 24-hour rolling mean of temperature), help in identifying trends and variability. Additionally, a monsoon phase parameter is introduced as a categorical feature to classify time periods into pre-monsoon, monsoon, and post-monsoon phases, reflecting seasonal climatic variations specific to Mumbai.

The parameters are used to train and evaluate multiple machine learning models (including XGBoost, LightGBM, Random Forest, and Deep Learning architectures) to predict WBGT and classify Heat Stress Levels based on established safety thresholds.

3.2 Data Format

IMD Dataset

The collected dataset is stored in CSV (Comma-Separated Values) format, which is one of the most widely used and versatile formats for handling structured data in scientific and analytical studies.

Advantages of Using CSV Format

- **1. Easy to Process and Analyze**: CSV files are simple and do not require complex parsing. They can be quickly loaded and processed using programming languages such as Python, where libraries like Pandas allow efficient data manipulation, cleaning, and analysis. This makes CSV ideal for handling large meteorological datasets.

- **2. High Compatibility Across Tools**: One of the biggest advantages of CSV is its universal compatibility. It can be seamlessly opened and used in:

  - Spreadsheet software like Microsoft Excel
  - Programming environments such as Jupyter Notebook
  - Machine learning frameworks like TensorFlow and Scikit-learn

  This flexibility allows the dataset to be easily transferred between different stages of analysis, from preprocessing to modeling and visualization.

- **3. Structured Tabular Representation**: CSV inherently organizes data in a clean tabular format, which is crucial for analytical workflows:
  - Rows = individual time-based observations (e.g., hourly or daily weather data)
  - Columns = variables/features (e.g., temperature, rainfall, wind components)

  This structure aligns perfectly with machine learning requirements, where data must be arranged in feature matrices for training models.

- **4. Lightweight and Efficient Storage**: Unlike complex binary formats, CSV files are lightweight and storage-efficient, making them easy to share, upload, and version-control. This is particularly useful when working with long-term climate datasets.

- **5. Transparency and Readability**: Since CSV is a plain-text format, it is:
  - Human-readable (can be opened in any text editor)
  - Easy to debug and verify
  - Less prone to compatibility issues compared to proprietary formats

ERA5 Dataset
The collected dataset is stored in CSV (Comma Separated Values) format. While the original source of the data is the ERA5 reanalysis (which is natively stored in GRIB or NetCDF formats), these specific files were exported as a spreadsheet-compatible version (.xlsx) and subsequently converted into the following two CSV files:

1. **ERA5_Daily_Mumbai.csv**: A structured data table containing daily meteorological observations such as maximum temperature (Tmax), wind speed, humidity, and other variables, indexed by date.

2. **Metadata.csv**: A reference file containing parameter definitions, measurement units (e.g., °C, m/s, hPa), and source information (Open-Meteo API).

3.3 Data Attributes

The dataset used in this study is highly comprehensive, consisting of a wide range of meteorological, environmental, and air quality parameters. These variables collectively enable a detailed analysis of heat stress, atmospheric dynamics, and environmental interactions in an urban setting like Mumbai. Each group of parameters plays a specific role in understanding both individual and combined effects on human thermal comfort and climate behavior.

1. Temporal Attributes (Time-Based Features) These variables define the time dimension of the dataset, which is essential for time-series analysis and trend identification.

- INDEX – Unique identifier for each observation

- YEAR, MN (Month), DT (Date), HR (Hour) – Structured time components

- DATETIME_IST / DATETIME_UTC – Timestamp in Indian Standard Time and Coordinated Universal Time

Importance:

- Enables analysis of diurnal (hourly) and seasonal patterns
- Helps detect heatwaves and extreme weather events
- Allows synchronization with global datasets (UTC)

2. Atmospheric Pressure Parameters

- SLP (Station Level Pressure)
- MSLP (Mean Sea Level Pressure)

Importance:

- Helps identify weather systems such as cyclones and depressions
- Influences wind patterns and atmospheric stability
- Useful for understanding large-scale climate behavior

3. Temperature Parameters

- DBT (Dry Bulb Temperature)
- WBT (Wet Bulb Temperature)
- DPT (Dew Point Temperature)
- AT (Apparent Temperature)

Importance:

- Core variables for heat stress analysis
- Wet bulb temperature is critical for human survivability thresholds
- Apparent temperature reflects perceived human discomfort

4. Humidity and Moisture Parameters

- RH (Relative Humidity)
- VP (Vapor Pressure)

Importance:

- High humidity reduces body cooling via sweating
- Contributes to heat index and discomfort

- Plays a role in cloud formation and precipitation

5. Wind Parameters

- DD / WD (Wind Direction)
- FFF / WS (Wind Speed)

Importance:

- Enhances heat dissipation from the human body
- Influences pollution dispersion
- Affects local weather and heat accumulation

6. Cloud and Weather Conditions

- VV (Visibility), AW (Present Weather)
- Cl, Cm, Ch – Low, medium, and high cloud types
- A, A.1, A.2 – Cloud amount at different layers
- Dl, Dm, Dh – Heights of cloud layers

Importance:

- Clouds regulate incoming solar radiation
- Affect day-night temperature variations
- Visibility indicates pollution and atmospheric clarity

7. Radiation and Heat-Related Parameters

- TC (Total Cloud Cover)
- H, Ht – Related to solar radiation or heat flux

Importance:

- Determines solar energy reaching the surface
- Impacts urban heat buildup
- Important for heat stress modeling

8. Hydrological Parameters

- RF (Rainfall)
- EVP (Evaporation)

- WAT (Water-related parameter)

Importance:

- Rainfall cools the environment and affects humidity
- Evaporation contributes to cooling
- Helps understand water cycle interactions

9. Additional Parameters

- DW, P, c, a – Auxiliary or coded meteorological variables
- Station_Code, Station_Name – Identify data source

Importance:

- Enables station-wise comparisons
- Supports filtering and grouping of data
- Useful for model training and feature engineering

3.3.1 Overall Significance of the Dataset

This dataset is highly comprehensive and multi-dimensional, enabling:

- Integrated heat stress analysis (temperature, humidity, wind)
- Urban climate studies (coastal vs inland variations)
- Machine learning-based prediction and forecasting

3.4 Data Integration and Merging Process

Since the data was collected from multiple time periods and not as a single dataset, a merging and consolidation process was required to create a unified dataset for analysis.

3.4.1 Merging of Yearly Datasets

The datasets corresponding to different time ranges:

- 1969–2009
- 2010–2024
- 2010–2025

were combined to form a continuous dataset. This merging process ensured that all available records from different time periods were integrated into a single structured dataset.

3.4.2 Creation of Combined Dataset (1969–2025)

After merging, the final dataset covered the full time period from 1969 to 2025. This combined dataset was used for all subsequent steps including imputation, feature analysis, and class generation.

The purpose of this consolidation was to:

- remove fragmentation of data across different files,
- enable long-term analysis,
- ensure consistency in further processing.

3.4.3 Handling Overlapping and Partial Periods

During the merging process, some datasets contained overlapping time ranges (for example, data from 2010–2024 and 2010–2025). These overlaps were handled carefully to avoid duplication and inconsistencies.

The merging process ensured that:

- duplicate entries were not introduced,
- partial data segments were properly aligned,
- the final dataset remained continuous and consistent across all years.

This step was crucial because improper merging could lead to incorrect analysis results, especially in later stages such as imputation and class generation.

3.4.4 ERA5 Dataset Consolidation

In contrast to the IMD dataset, the ERA5 dataset is provided as a single, pre-structured file, eliminating the need for merging multiple yearly datasets. The data was directly loaded into the processing environment (Google Colab) from an Excel/CSV source.

The consolidation process primarily involved data standardization and structuring rather than merging. Initially, all column names were normalized by removing unit annotations and ensuring consistent naming conventions (for example, converting "Tmax (°C)" to "Tmax"). This step ensured compatibility with downstream processing and feature engineering pipelines.

The date column was identified and parsed into a proper datetime format, allowing accurate temporal indexing. Following this, the dataset was sorted in chronological order from 1 January 2015 to 9 March 2026 to maintain temporal consistency, which is essential for time-series modeling.

Additionally, basic integrity checks were performed to ensure:

- no duplicate records,
- consistent time intervals (daily frequency),
- proper alignment of all variables.

This structured and chronologically ordered dataset forms the base input for further analysis, feature engineering, and predictive modeling in the ERA5 pipeline.

3.5 Data Quality Check and Missing-Value Analysis

After merging the datasets from different time periods and stations, a detailed data quality check was performed to ensure the reliability and consistency of the dataset before further processing. Since the data was collected over a long time span and from multiple sources, the presence of missing and incomplete values was expected. Therefore, identifying and analysing missing data was a crucial step in the methodology.

3.5.1 Null / Missing Value Identification

The first step in data quality assessment involved identifying missing or null values in the dataset. Each variable in the dataset was examined to detect the presence of missing entries.

This was done to:

- understand the extent of incomplete data,
- identify which variables were affected the most,
- determine the need for appropriate imputation techniques.

The identification process ensured that all missing values across the dataset were clearly marked and prepared for further analysis and treatment.

3.5.2 Percentage of Missingness

Once the missing values were identified, the percentage of missingness was calculated for the dataset. This involved determining the proportion of missing values relative to the total number of observations.

The calculation of missingness percentage helped in:

- quantifying the severity of missing data,
- comparing the level of data incompleteness across different parts of the dataset,
- guiding the selection of suitable imputation methods.

This step provided a clear understanding of how much of the dataset required reconstruction through imputation techniques.

3.5.3 Station-wise and Variable-wise Missingness

To gain deeper insights into the missing data patterns, the analysis was further extended in two dimensions:

1. Station-wise Missingness

Missing values were analyzed separately for:

- Colaba
- Santacruz

This helped in identifying whether one station had more missing data than the other and understanding differences in data quality between the two locations.

2. Variable-wise Missingness

Missing values were also analyzed for each variable individually. This allowed the identification of:

- variables with high missingness,
- variables with relatively complete data,
- variables that required more advanced imputation techniques.

3.5.4 ERA5 Dataset Data Quality

For the ERA5 dataset, data quality checks were performed after loading to ensure consistency and reliability. Since ERA5 is a reanalysis dataset, it is inherently complete and physically consistent, resulting in minimal missing values in the raw data.

Basic validation included checking:

- data types and value ranges,
- duplicate records,
- consistent daily time intervals.

The initial dataset showed negligible missingness. However, during feature engineering, NaN values were introduced due to:

- lag features,
- rolling window calculations,
- derived features.

These NaNs occur at the beginning of the time series and were handled by removing or aligning incomplete records.

Overall, ERA5 required minimal missing-value handling compared to the IMD dataset.

3.5.5 Data Imputation

Data imputation is the process of replacing missing values in the dataset with estimated values to obtain a complete and consistent dataset for further analysis. Since the collected IMD data contained missing entries across different time periods and stations, multiple imputation techniques were applied and compared.

The objective of applying multiple methods was to:

- Handle different patterns of missing data
- Improve data completeness
- Identify the most suitable imputation technique based on performance

The imputation methods were evaluated using the following performance metrics:

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of Determination (R^2)

Lower values of MSE, RMSE, and MAE indicate better performance, while higher R^2 indicates better accuracy.

3.5.5.1 Basic Statistical Imputation Methods

Mean Imputation

Missing values are replaced with the mean of the observed values:

$$\hat{x}_{\text{missing}} = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \tag{3.1}$$

Median Imputation

Missing values are replaced with the median of the observed values:

$$\hat{x}_{\text{missing}} = \text{median}(x_1, x_2, \ldots, x_n) \tag{3.2}$$

This method is less sensitive to outliers compared to mean imputation.

3.5.5.2 Time-Series Based Imputation

Linear Interpolation

Missing values are estimated using neighboring values in the time sequence:

$$\hat{x}_t = x_{t_1} + \frac{t - t_1}{t_2 - t_1} \cdot (x_{t_2} - x_{t_1}) \tag{3.3}$$

This method is suitable when data follows a continuous trend.

3.5.5.3 Machine Learning-Based Imputation

K-Nearest Neighbors (KNN) Imputation

Missing values are estimated using the average of the nearest neighbors:

$$\hat{x}_{\text{missing}} = \frac{1}{k}\sum_{j=1}^{k} x_j^{(\text{neighbor})} \tag{3.4}$$

Multivariate Imputation by Chained Equations (MICE)

Each variable is iteratively predicted using other variables:

$$X_j = f(X_1, X_2, \ldots, X_{j-1}, X_{j+1}, \ldots, X_p) + \epsilon \tag{3.5}$$

3.5.5.4 Hybrid Imputation Methods

KNN-MICE Weighted Imputation

$$\hat{x} = 0.5 \cdot \hat{x}_{\text{KNN}} + 0.5 \cdot \hat{x}_{\text{MICE}} \tag{3.6}$$

Average KNN-Mean-Interpolation

$$\hat{x} = \frac{\hat{x}_{\text{KNN}} + \hat{x}_{\text{Mean}} + \hat{x}_{\text{Interp}}}{3} \tag{3.7}$$

Weighted KNN-Mean-Interpolation

$$\hat{x} = 0.5 \cdot \hat{x}_{\text{KNN}} + 0.2 \cdot \hat{x}_{\text{Mean}} + 0.3 \cdot \hat{x}_{\text{Interp}} \tag{3.8}$$

Dynamic Weighted Hybrid Method

$$\hat{x} = w_{\text{mean}} \cdot \hat{x}_{\text{Mean}} + w_{\text{interp}} \cdot \hat{x}_{\text{Interp}} + w_{\text{KNN}} \cdot \hat{x}_{\text{KNN}} + w_{\text{MICE}} \cdot \hat{x}_{\text{MICE}} \tag{3.9}$$

Weights are adjusted based on local data availability.

Column-wise Weighted Hybrid Method

$$w_m^{(c)} = \frac{1/\text{RMSE}_m^{(c)}}{\sum_{m'} 1/\text{RMSE}_{m'}^{(c)}} \tag{3.10}$$

$$\hat{x}^{(c)} = w_{\text{mean}}^{(c)} \cdot \hat{x}_{\text{Mean}} + w_{\text{interp}}^{(c)} \cdot \hat{x}_{\text{Interp}} + w_{\text{KNN}}^{(c)} \cdot \hat{x}_{\text{KNN}} + w_{\text{MICE}}^{(c)} \cdot \hat{x}_{\text{MICE}} \tag{3.11}$$

Each variable gets its own optimized weights.

3.5.5.5 ERA5 Dataset Handling

For the ERA5 dataset, no separate or extensive imputation study was required. As a reanalysis dataset, ERA5 is inherently complete and contains minimal missing values in the raw data.

Only basic preprocessing steps were applied where necessary. Missing values that appeared during feature engineering (such as lag and rolling features) were handled through standard cleanup procedures, including removal or alignment of incomplete initial records.

Thus, unlike the IMD dataset, the ERA5 pipeline does not rely on multiple imputation techniques and focuses primarily on feature generation and modeling.

Chapter 4
Proposed Methodology

4.1 Methodology

This project aims to develop a comprehensive and data-driven framework for analyzing and predicting heat stress conditions in Mumbai. It focuses on integrating long-term historical meteorological data with modern reanalysis datasets to capture both past trends and future patterns of heat stress. Advanced machine learning and deep learning techniques are applied to improve prediction accuracy while maintaining model interpretability. The objectives are structured to address key challenges such as data quality, class imbalance, and temporal dependencies, ultimately contributing to reliable and practical heat stress assessment for real-world applications.

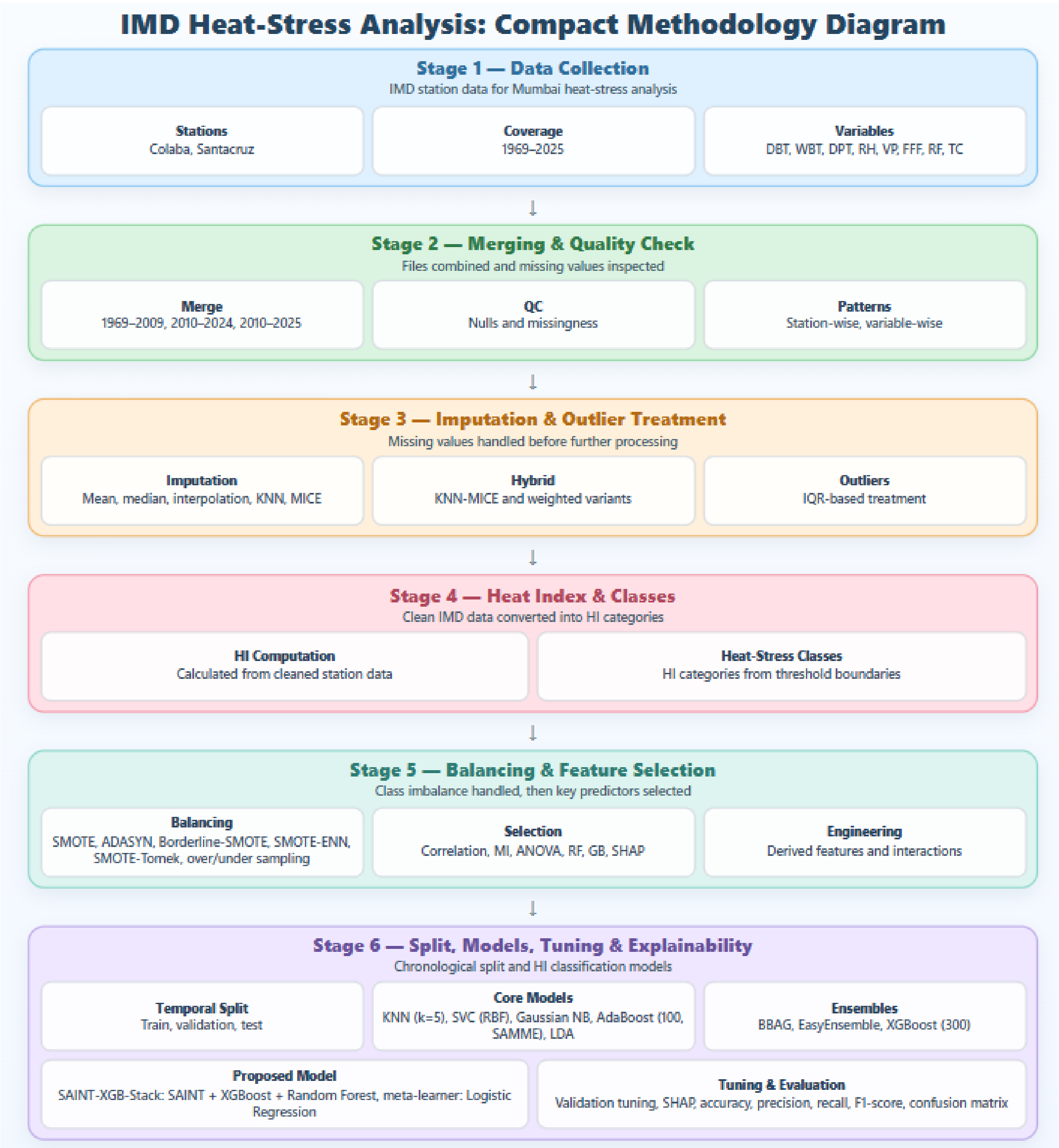

Figure 4.1: IMD Methodology Overview

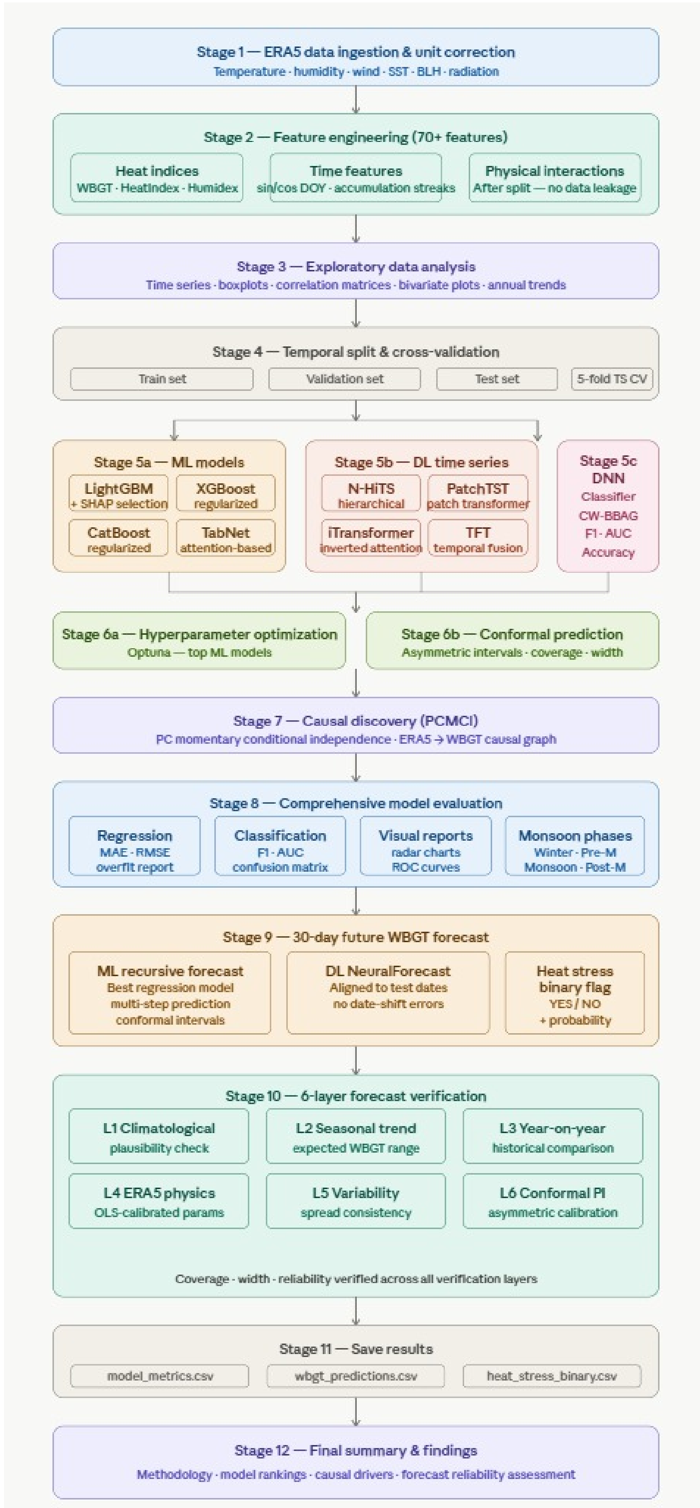

Figure 4.2: ERA5 Methodology Overview

4.2 Overall System Workflow

The project follows a structured dual-track workflow to analyze and predict heat stress conditions in Mumbai by combining historical data analysis with modern predictive modeling techniques. This approach ensures both long-term trend understanding and reliable future forecasting.

The first part utilizes data from the India Meteorological Department, where station-based synoptic observations are processed to study historical heat stress patterns using the Heat Index (HI). This involves data consolidation, preprocessing, handling missing values, and generating heat stress classifications.

The second part uses reanalysis-based meteorological data to model and predict Wet-Bulb Globe Temperature (WBGT). This includes feature engineering, temporal data splitting, model training, and evaluation using multiple machine learning and deep learning techniques. The overall workflow can be summarized as:

1. **Data Collection and Preparation:** Collection and preparation of IMD and reanalysis datasets.
2. **Data Preprocessing and Quality Control:** Cleaning of data, handling missing values, and ensuring consistency.
3. **Computation of Heat Stress Indicators:** Calculation of heat stress indices such as Heat Index (HI) and Wet Bulb Globe Temperature (WBGT).
4. **Feature Engineering and Dataset Structuring:** Extraction of relevant features and organization of datasets for modeling.
5. **Model Development and Evaluation:** Training and testing of predictive models.
6. **Forecasting and Validation:** Generation of forecasts and validation of predicted heat stress.

This integrated framework enables a comprehensive understanding of heat stress by combining historical analysis with predictive modeling, making it suitable for both climate studies and early warning applications.

4.3 Study Data and Problem Setup

The study uses two datasets to analyze and predict heat stress in Mumbai. The first dataset, obtained from the India Meteorological Department, consists of long-term synoptic hourly observations (1969–2025) from Santacruz and Colaba stations, including variables such as temperature, humidity, wind speed, and vapor pressure. This dataset is used to compute the Heat Index (HI) for historical heat stress analysis.

The second dataset consists of daily reanalysis data (2015–2026) with 4,086 observations and variables such as Tmax, relative humidity, wind components, radiation, and pressure. This dataset is used to predict Wet-Bulb Globe Temperature (WBGT), formulated as both a regression problem (continuous prediction) and a classification problem (heat stress detection).

4.4 Data Preprocessing and Quality Control

The station-based dataset required merging multiple time periods, aligning time records, removing duplicate entries, and identifying missing values due to its long historical span. The reanalysis dataset required column normalization, date parsing, chronological sorting, and unit consistency checks. Missing values were minimal initially but appeared during feature engineering steps such as lag and rolling calculations. Care was also taken to avoid data leakage by excluding variables directly related to the target variable.

4.5 IMD Data Consolidation, Imputation, and Outlier Treatment

The station-based dataset was merged into a continuous time series and cleaned by removing duplicate records. Missing values were handled using multiple imputation techniques, including statistical, interpolation, and machine learning-based methods. Outliers were treated using the Interquartile Range (IQR) method, and imputation performance was evaluated both with and without outlier handling. The methods were compared using MSE, RMSE, MAE, and R^2 score, and the best-performing approach was selected for final dataset preparation.

4.6 ERA5 Variable Preparation and Derived Feature Setup

The ERA5 dataset was preprocessed by standardizing units and ensuring consistency across all meteorological variables. Derived variables such as wind speed, radiation-based features, and thermodynamic parameters were computed from the base inputs. Additional features were generated to enhance model performance, while variables directly related to the target were excluded to prevent data leakage. The final feature set was then structured and prepared for model training and evaluation.

4.7 Heat-Stress Variable Computation

For the IMD dataset, heat stress was quantified using the Heat Index (HI), which combines air temperature and relative humidity to represent the perceived temperature experienced by the human body. The HI values were computed from the cleaned synoptic observations and used to categorize different levels of heat stress for long-term historical analysis.

For the ERA5 dataset, Wet-Bulb Globe Temperature (WBGT) was computed as the primary target variable using multiple meteorological inputs, including temperature, humidity, wind, and radiation. WBGT was explicitly treated as the main prediction variable in the framework due to its strong physiological relevance. Other heat-related indices were generated only as secondary or comparison variables and were not used as primary targets in the modeling process.

The WBGT was calculated using the following formulation:

$$
\begin{aligned}
WBGT &= 0.7 \times T_{nwb} + 0.2 \times T_g + 0.1 \times T \\
T_g &= T + 0.0256 \times SR - 0.5 \\
T_{nwb} &= T \cdot \arctan\!\Big(0.151977 \times \sqrt{RH + 8.313659}\Big) \\
&\quad + \arctan(T + RH) - \arctan(RH - 1.676331) \\
&\quad + 0.00391838 \times RH^{1.5} \times \arctan(0.023101 \times RH) - 4.686035
\end{aligned}
$$

where T is air temperature (°C), RH is relative humidity (%), and SR represents solar radiation.

4.8 Heat-Stress Classification and Balancing

In the case of the IMD dataset, Heat Index (HI) values were transformed into categorical heat-stress levels to support classification-based analysis. As extreme heat events were relatively rare in this long-term record, class imbalance was addressed using multiple resampling techniques to improve model performance. The methods explored include SMOTE, ADASYN, Borderline-SMOTE, SMOTE-ENN, SMOTE-Tomek, along with random oversampling and random undersampling. Among these, SMOTE-ENN was found to provide a well-balanced dataset while also reducing noise.

For the ERA5-based analysis, WBGT values were categorized into standard heat-stress levels such as Normal, Caution, Danger, and Extreme, along with a binary HeatStress label for classification tasks. Instead of applying resampling techniques, imbalance was handled through:

1. **Temporal Splitting:** Data is divided based on time to preserve real-world distribution and avoid data leakage.

2. **Model-based Weighting and Evaluation:** Application of weighting techniques and appropriate evaluation metrics to improve model performance and handle class imbalance.

This approach ensures realistic performance assessment while maintaining the temporal structure of the data.

4.9 Feature Engineering and Feature Selection

Feature engineering and selection were performed to enhance model performance and identify the most relevant variables driving heat stress.

In the case of the IMD dataset, the focus was on selecting the most influential features affecting the Heat Index (HI). A combination of statistical and machine learning-based techniques was used to ensure robust and unbiased selection. These included: Feature Selection Techniques:

- Correlation analysis
- Mutual Information
- ANOVA F-test

- Random Forest importance
- Gradient Boosting importance
- SHAP-based importance

For the ERA5 dataset, the emphasis was on constructing a comprehensive set of derived features to capture temporal patterns and underlying physical relationships. This resulted in a rich and diverse feature space aimed at improving predictive capability. The engineered features include:

- Lag features to capture past dependencies
- Rolling statistics (mean, max, min, standard deviation)
- Anomaly features based on recent deviations
- Physical interaction terms between variables
- Short-term trend features
- Seasonal and monsoon phase encodings

Additionally, variables that could introduce data leakage were excluded to ensure that the models learn true atmospheric drivers rather than direct proxies of the target.

4.10 Temporal Splitting and Validation Design

In the IMD dataset, a chronological splitting approach was used where the data was divided into training, validation, and testing sets based on year. This ensures that the model is trained on past observations and evaluated on future data, preserving the natural time order and avoiding data leakage.

In the ERA5 dataset, a more advanced temporal validation strategy was applied using a walk-forward approach with an expanding window. In this method, the model is repeatedly trained on earlier time periods and tested on the next unseen segment. Additionally, a 5-fold TimeSeries cross-validation strategy was used to improve robustness. This ensures that evaluation is performed under realistic forecasting conditions while avoiding lookahead bias.

4.11 Explainability and Causal Discovery

In the IMD dataset, explainability was achieved using multiple feature-ranking techniques to understand the influence of meteorological variables on Heat Index (HI). Methods such as correlation, mutual information, ANOVA, and tree-based importance were used, along with SHAP to provide both global and local interpretability. In the ERA5 dataset, explainability was mainly performed using SHAP analysis to determine global and phase-wise feature importance, helping identify the key variables influencing WBGT. Additionally, a causal discovery approach using PCMCI (Peter and Clark Momentary Conditional Independence) was applied to identify true drivers of WBGT. This method captures time-lagged and conditional relationships, ensuring that the identified factors represent meaningful causal influences rather than simple correlations.

Chapter 5

Implementation

5.1 Model Development and Training

The model development stage is implemented separately for the two parts of the project. The station-based pipeline focuses on Heat Index (HI) classification using a stacked ensemble, while the ERA5 pipeline focuses on WBGT prediction and heat-stress detection using a diverse set of machine learning and deep learning models. In both cases, the objective is to develop models that are accurate, stable, and capable of generalizing well under temporal validation.

5.1.1 Station-Based Model Development (IMD)

The IMD-based implementation uses a stacked ensemble architecture, where multiple models are combined to improve prediction performance. The base learners include SAINT (Self-Attention and Intersample Transformer), XGBoost (Extreme Gradient Boosting), and Random Forest, while Logistic Regression is used as a meta-learner to combine their outputs. These models are integrated within a unified framework to enhance overall prediction accuracy and robustness. The stacked ensemble approach allows the system to leverage the strengths of multiple learning algorithms and improve generalization across different data patterns. The final prediction is obtained by combining the outputs of SAINT, XGBoost, and Random Forest using Logistic Regression.

- SAINT (Self-Attention and Intersample Transformer): A transformer-based model designed for tabular data that captures complex feature interactions using attention mechanisms.

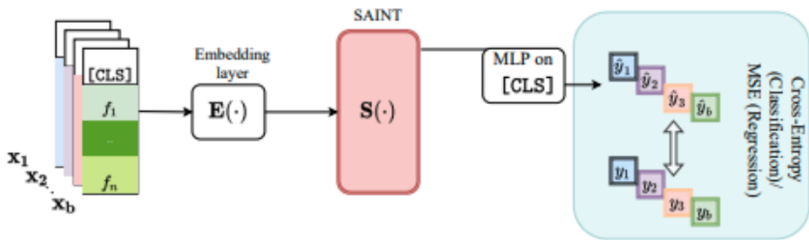

Figure 5.1: SAINT Architecture

- XGBoost (Extreme Gradient Boosting): A tree-based ensemble method that builds models sequentially and effectively captures non-linear relationships.
- Random Forest: A bagging-based ensemble method that improves model stability and generalization by using multiple decision trees.

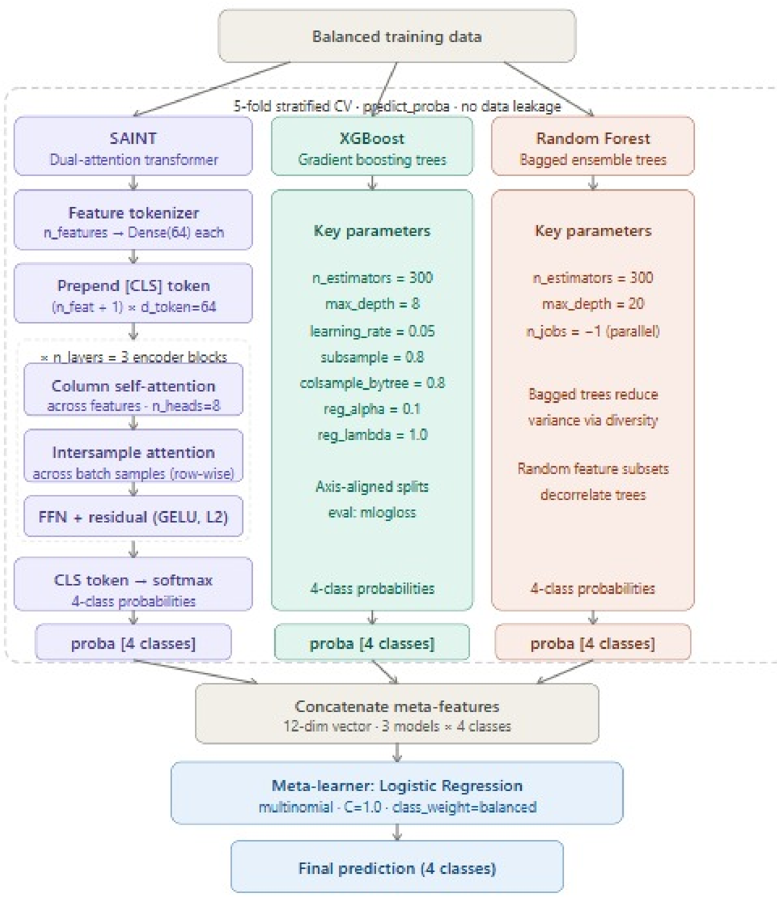

Figure 5.2: SAINT + XGBoost + Random Forest Architecture

- Logistic Regression: Acts as the meta-learner in the stacking framework, combining predictions from the base models into a final output.

5.1.2 ERA5-Based Model Development

The ERA5 pipeline implements a unified framework that compares multiple machine learning and deep learning models for WBGT prediction. These models are designed to capture both static relationships and temporal dependencies within the data. The main models used include:

- LightGBM: A gradient boosting framework that uses histogram-based learning for faster training. It efficiently handles large datasets and captures complex non-linear relationships with strong regularization support.

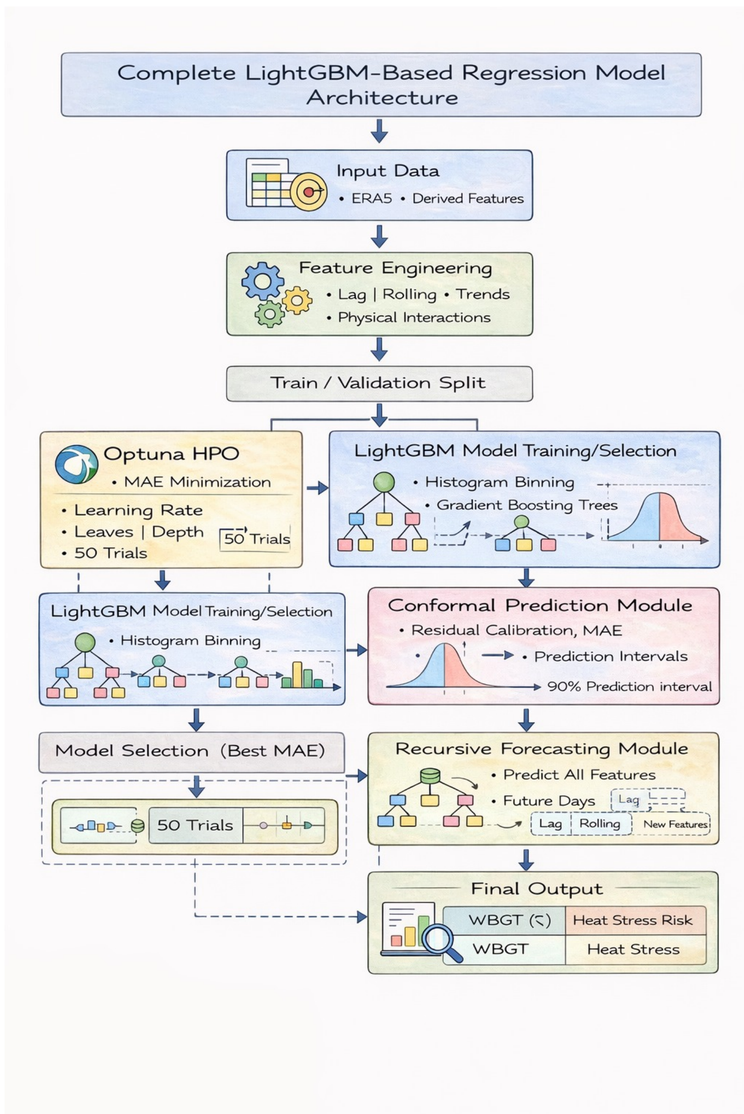

Figure 5.3: LightGBM Architecture

- XGBoost: A gradient boosting method known for its robustness, regularization techniques, and ability to handle structured data effectively.

- CatBoost: A boosting algorithm that handles categorical and numerical data efficiently while reducing overfitting through ordered boosting techniques.

- TabNet: A deep learning model designed for tabular data, which uses sequential attention to select important features at each decision step.

- DNN-CW-BBAG: A custom deep neural network ensemble using class-weighted balanced bagging, where multiple neural networks are trained on balanced subsets and combined using a meta-learning approach.

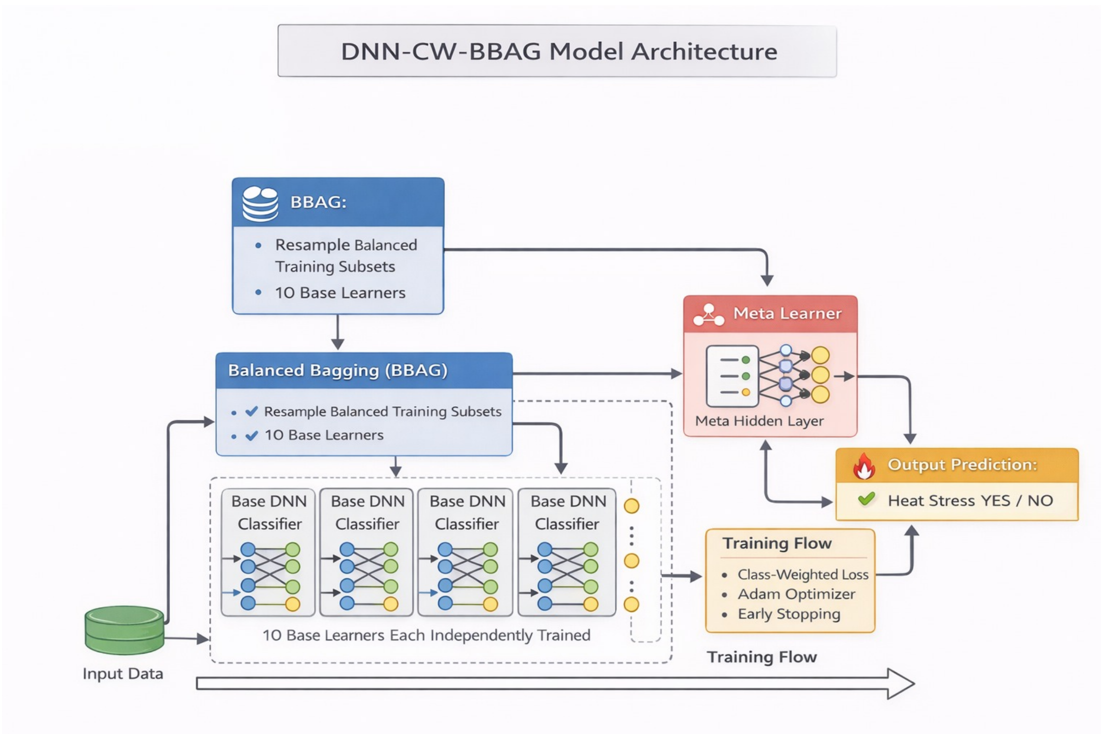

Figure 5.4: DNN-CW-BBAG Architecture

- N-HiTS: A neural forecasting model for time series prediction that captures temporal patterns using hierarchical interpolation.

- PatchTST: A transformer-based time series model that processes input data in patches, enabling efficient learning of long-range dependencies.

- iTransformer: An improved transformer architecture tailored for time series data, focusing on better feature representation and sequence modeling.

- TFT (Temporal Fusion Transformer): A hybrid model combining recurrent networks and attention mechanisms to capture temporal dynamics and feature importance.

These models are trained under a consistent framework for both regression (continuous WBGT prediction) and classification (heat-stress detection using thresholds). Strong regularization techniques such as shallow trees, subsampling, and parameter tuning are applied to prevent overfitting. The comparative evaluation shows that tree-based boosting models, particularly tuned LightGBM, and the DNN-CW-BBAG ensemble achieve the best performance, while transformer-based models show comparatively lower performance in this specific setup.

5.2 Hyperparameter Optimization and Regularization

In the IMD-based implementation, the proposed SAINT-XGB-Stack model was built using SAINT, XGBoost, and Random Forest as base learners, with Logistic Regression as the meta-learner. A 5-fold cross-validated stacking strategy was used to generate meta-features without data leakage, which helped improve generalization and reduce overfitting risk. The model achieved strong temporal performance, with the stacked ensemble outperforming the individual base models on the test set. In the ERA5 pipeline, hyperparameter optimization was carried out using Optuna with the TPE sampler. The tuning process focused on improving model performance while controlling complexity through regularization and suitable tree settings. Strong overfitting control was maintained using a temporal train-validation-test split, expanding-window cross-validation, leakage-free feature engineering, and early stopping with Batch Normalization and Dropout in deep learning models.

5.3 Recursive 30-Day WBGT Forecasting

For the final forecasting stage, the best regression model, LightGBM-Tuned, was used to generate a 30-day ahead WBGT forecast through a recursive multi-step approach. In this method, the model predicted one future day at a time, and each predicted WBGT value was written back into the growing forecast sequence so that the next day could be estimated using the updated lag and temporal features. This allowed the model to preserve the time-dependent structure of the ERA5 feature set during long-horizon forecasting. The forecast was generated from the last known date in the dataset and extended for 30 days ahead using the selected tuned model. The final forecast showed that 27 out of 30 days were classified as heat stress days, with WBGT values remaining mostly in the high-risk range. The forecast summary also reported a mean WBGT of about 30.8°C, with values ranging from roughly 29.9°C to 31.9°C, confirming persistent heat-stress conditions over the prediction horizon.

5.4 Forecast Verification and Reliability Assessment

To ensure that the 30-day WBGT forecast was not only accurate but also physically and statistically reliable, a six-layer verification framework was applied after the LightGBM-Tuned recursive forecast was generated. This stage was used to test whether the predicted values were consistent with historical climatology, seasonal behaviour, year-to-year patterns, ERA5-based physical relationships, and the natural variability of the WBGT series. The forecast was also verified using an asymmetric conformal prediction interval, which provided calibrated uncertainty bounds with a target historical coverage of at least 80The first layer checked climatological plausibility by comparing each forecast day with the historical day-of-year mean and standard deviation. The second layer examined the seasonal warming trend expected from March to April. The third layer performed a year-on-year comparison against the same calendar period in previous years. The fourth layer tested physics consistency by comparing the forecast with a calibrated ERA5-based WBGT formulation using solar radiation. The fifth layer assessed forecast variability and lag-1 autocorrelation to ensure the series was neither unrealistically smooth nor excessively noisy. The final layer used asymmetric conformal prediction with bias correction and interval scaling to produce reliable prediction intervals, and the final verification summary reported a complete pass across all six layers, indicating that the forecast was reliable.

5.5 Final Model Comparison and Output Generation

In the IMD-based implementation, the final comparison showed that the stacked ensemble model provided the best overall classification performance. The outputs from SAINT, XGBoost, and Random Forest were combined using Logistic Regression, and the final test results confirmed that the ensemble was more reliable than the individual models. The trained model outputs and comparison results were saved for further analysis. In the ERA5 pipeline, the final comparison of all trained models showed that DNN-CW-BBAG achieved the best classification performance, while LightGBM-Tuned gave the best regression results for WBGT prediction. The final outputs included the model comparison table, saved predictions, and the 30-day recursive WBGT forecast used for verification and analysis.

Chapter 6

Results & Discussion

6.1 ERA5 Dataset

6.1.1 Model Training Results

The proposed system utilizes supervised machine learning models such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) to predict the Wet Bulb Globe Temperature (WBGT) using meteorological parameters derived from the ERA5 dataset. These models are capable of capturing complex nonlinear relationships between environmental variables such as temperature, humidity, wind speed, and solar radiation, making them suitable for heat stress prediction.

The dataset was divided into training and testing subsets to evaluate the generalization capability of the model. An 80:20 split was adopted, where 80% of the data was used for training the model and 20% was reserved for testing.

Table 6.1: Test Set Comparison of Regression Models

Model	**MAE**	**RMSE**	**F1**	**AUC**	**Precision**	**Recall**	**Accuracy**
LightGBM	0.2329	0.3112	0.9455	0.9919	0.8916	0.9487	0.9582
XGBoost	0.3037	0.3934	0.9352	0.9893	0.8743	0.9359	0.9502
CatBoost	0.2243	0.2874	0.9525	0.9945	0.8844	0.9808	0.9630
TabNet	0.7434	1.0290	0.8497	0.9310	0.8296	0.7179	0.8923
LightGBM-SHAP30	0.2540	0.3384	0.9394	0.9911	0.8802	0.9423	0.9534
LightGBM-Tuned	0.1294	0.1839	0.9707	0.9978	0.9277	0.9872	0.9775
N-HiTS	1.1507	1.4083	0.6923	0.8336	0.7808	0.3654	0.8151
PatchTST	1.3860	1.7148	0.7351	0.8704	0.8904	0.4167	0.8408
iTransformer	1.1801	1.4209	0.7673	0.8254	0.8736	0.4872	0.8537

Multiple machine learning and deep learning models were evaluated for predicting WBGT using ERA5 meteorological parameters. These models include LightGBM, XGBoost, Cat-

Boost, TabNet, and advanced time-series models such as N-HiTS, PatchTST, and iTransformer.

The performance of each model was assessed using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Accuracy. The results of the comparative analysis are presented in Table 6.1.

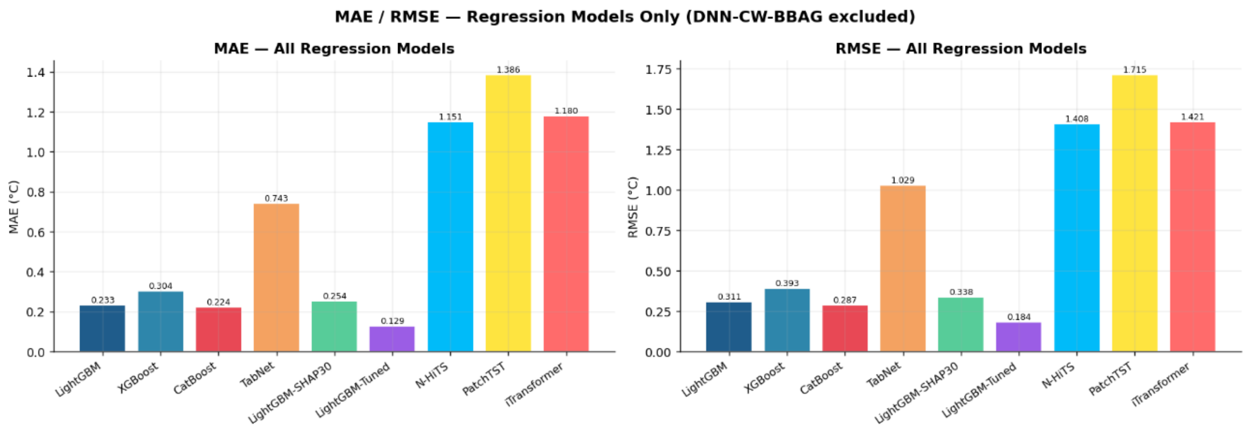

Figure 6.1: MAE & RMSE Bar Graph

The MAE and RMSE comparison illustrates that tree-based models such as LightGBM and CatBoost outperform deep learning models. The LightGBM-Tuned model achieves the lowest error values, confirming its effectiveness.

To assess the accuracy and reliability of the model, several performance evaluation metrics were used:

6.1.1.1 Mean Absolute Error (MAE)

MAE measures the average absolute difference between the predicted and actual WBGT values. Lower MAE indicates better model performance.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{6.1}$$

6.1.1.2 Root Mean Square Error (RMSE)

RMSE provides a measure of the magnitude of prediction errors, giving higher weight to larger errors. It is useful for understanding the model's sensitivity to outliers.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6.2}$$

6.1.1.3 Coefficient of Determination (R^2 Score)

The R^2 score represents the proportion of variance in the dependent variable (WBGT) that is predictable from the independent variables. A value closer to 1 indicates a better fit of the model.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{6.3}$$

The performance of the model was evaluated on both training and testing datasets. A small difference between training and testing errors indicates that the model generalizes well and does not suffer from overfitting. The obtained results demonstrate that the model achieves high predictive accuracy with minimal error, making it suitable for heat stress forecasting applications.

6.1.2 Model Evaluation

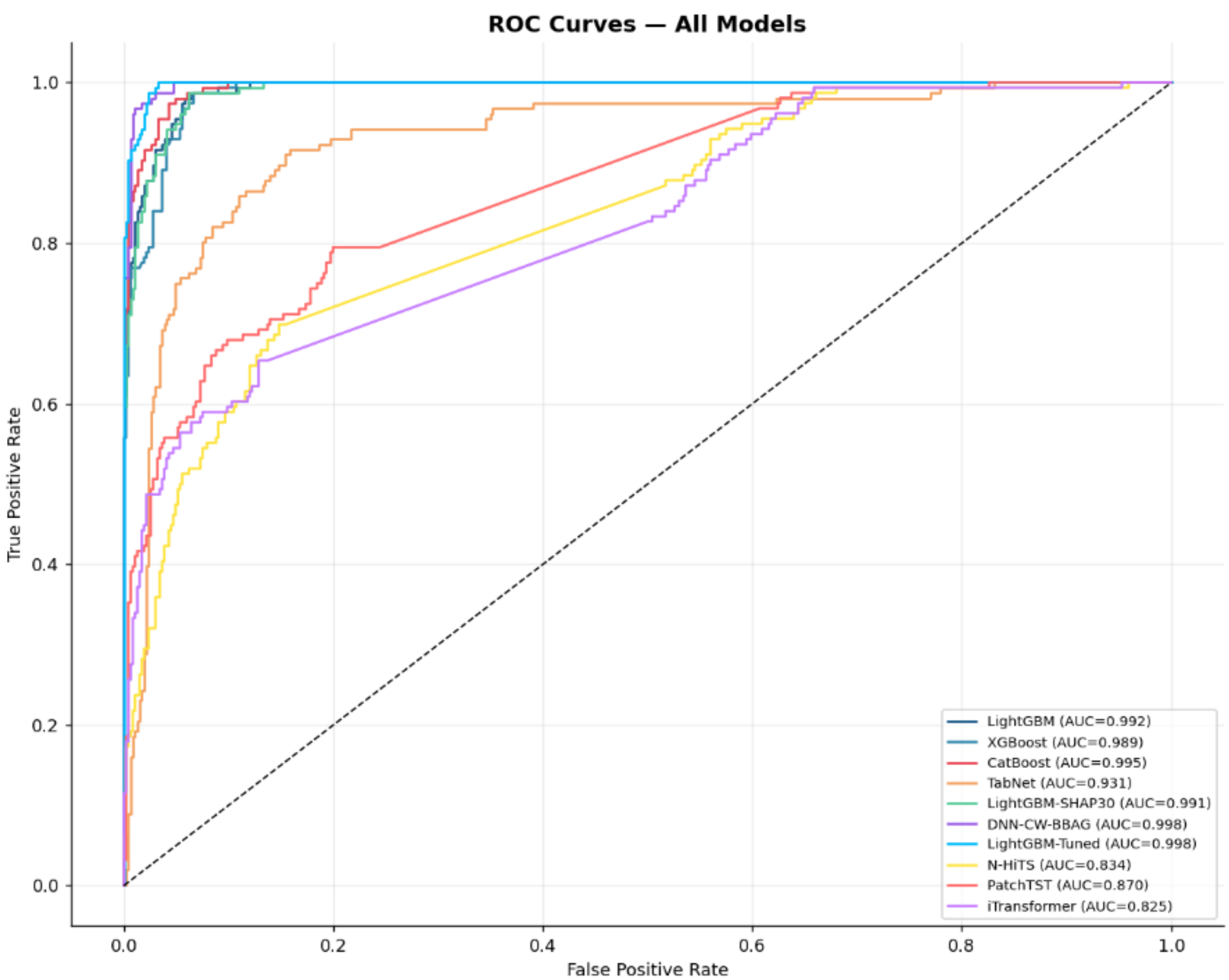

Figure 6.2: ROC Curve Comparison of All Models for WBGT Prediction

The ROC curves demonstrate that most models achieve high AUC values, indicating strong discriminative ability in classifying heat stress conditions. In particular, the LightGBM-Tuned and CatBoost models exhibit the highest AUC scores, with their curves closely approaching the top-left corner of the ROC space. This reflects a high true positive rate (sensitivity) while maintaining a low false positive rate, which is essential for reliable heat stress prediction. The superior performance of these models suggests their effectiveness in capturing complex relationships between meteorological variables and heat stress outcomes. Overall, the results confirm that the selected models are capable of accurately distinguishing between heat stress and non-heat stress conditions, making them suitable for real-world forecasting applications.

6.1.3 Forecasting Results & Analysis

The WBGT forecast for the next 30 days in Figure 6.3 shows a consistent increasing trend, with most values exceeding the heat stress threshold of 30°C. The shaded region indicates predicted heat stress days, highlighting potential risk periods.

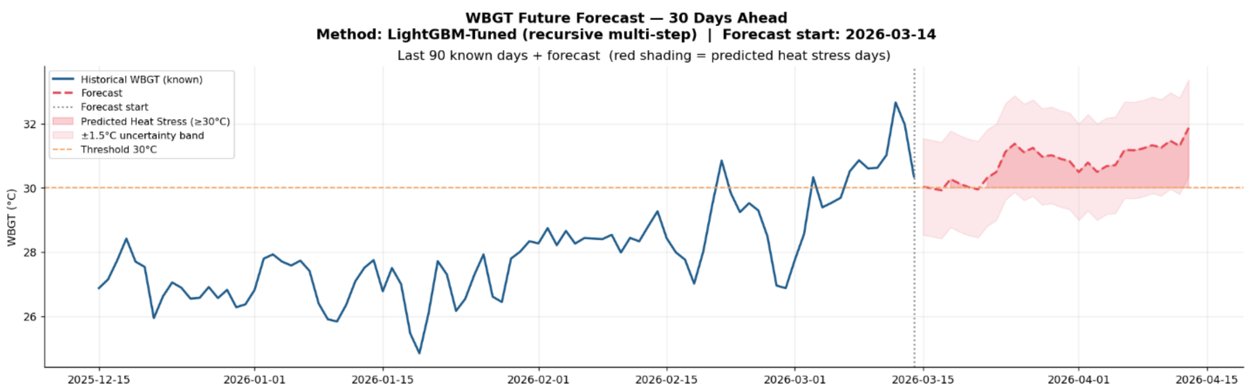

Figure 6.3: WBGT Future Forecast Graph

The daily forecast in Figure 6.4 indicates that 27 out of 30 days are predicted to experience heat stress conditions. This highlights the severity of heat stress in the studied region and emphasizes the need for preventive measures.

The forecast verification dashboard compares predicted WBGT values with historical climatological patterns. The forecast closely aligns with the historical mean while remaining consistently above the heat stress threshold. The inclusion of uncertainty bands demonstrates that the predictions are reliable and fall within acceptable confidence limits. This confirms the robustness and stability of the forecasting model.

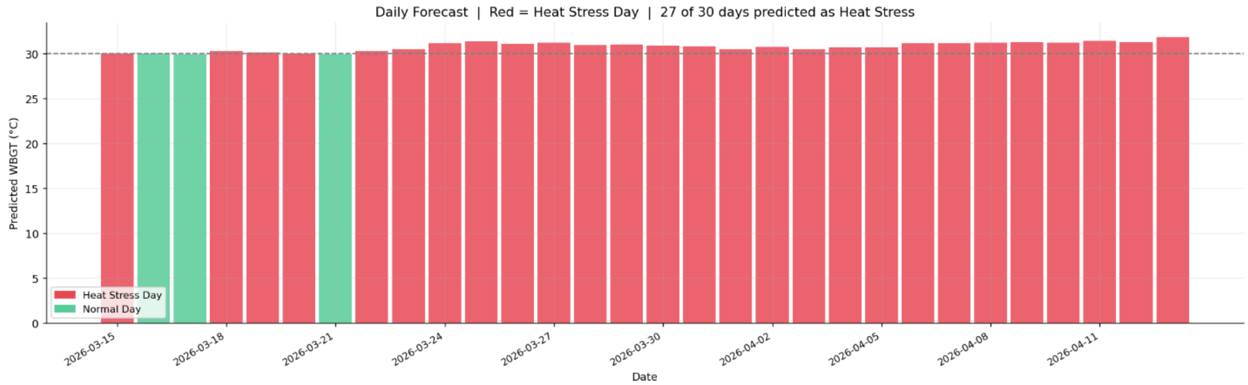

Figure 6.4: Heat Stress Days Bar Graph

The forecasting results indicate a strong likelihood of persistent heat stress conditions over the next 30 days. The model successfully captures temporal trends and seasonal behavior, while the verification analysis confirms the reliability of predictions. The high number of predicted heat stress days highlights potential risks to human health and emphasizes the importance of proactive measures such as heat warnings, hydration strategies, and urban cooling interventions.

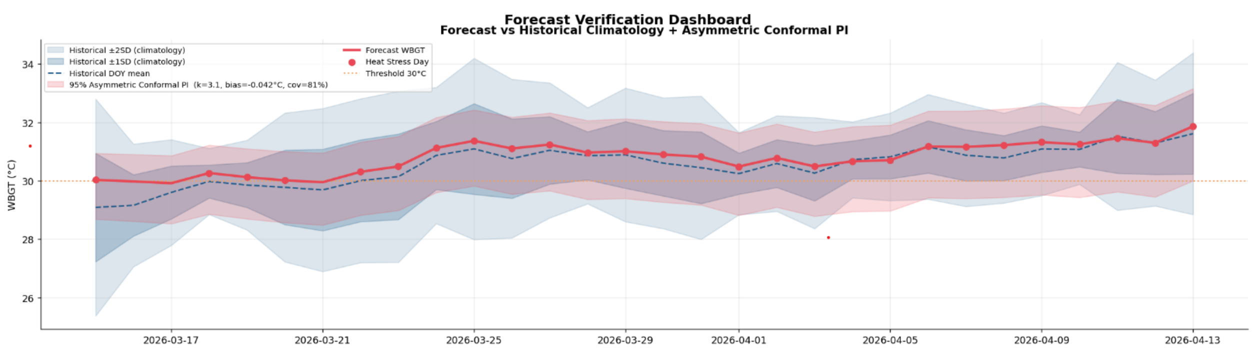

Figure 6.5: Forecast verification comparing predicted WBGT with historical climatology and uncertainty intervals

The final forecast results confirm that the model is reliable for predicting heat stress conditions. The LightGBM-Tuned model achieved low error values, with MAE of 0.129°C and RMSE of 0.184°C, indicating high prediction accuracy.

The forecast analysis shows that 27 out of 30 days fall under heat stress conditions, with WBGT values consistently exceeding the critical threshold of 30°C. This highlights the severity of heat exposure during the forecast period.

The low bias correction value (-0.042°C) and high prediction interval coverage (81.3%) demonstrate that the model produces stable and well-calibrated predictions. Overall, the

Table 6.2: Final Forecast Performance Summary

Parameter	**Value**
Model Used	LightGBM-Tuned
Forecast Period	15 March 2026 – 13 April 2026
MAE	0.129°C
RMSE	0.184°C
Heat Stress Days	27 / 30 days
WBGT Range	29.92°C – 31.87°C
Bias Correction	-0.042°C
Prediction Interval Coverage	81.3%

results validate the effectiveness of the proposed system for real-world heat stress forecasting.

6.2 IMD Dataset

6.2.1 Exploratory Data Analysis

The monthly Heat Index (HI) analysis in Figure 6.6 demonstrates a clear and consistent seasonal variation, with significantly higher values observed during the summer and monsoon periods. This trend is evident across both inland (Santacruz) and coastal (Colaba) regions, although the intensity varies due to geographical influences. During the summer months, rising air temperatures combined with increasing atmospheric moisture contribute to elevated HI levels, indicating heightened thermal discomfort. This effect becomes even more pronounced during the monsoon season, where persistently high humidity restricts evaporative cooling of the human body, thereby intensifying perceived heat stress.

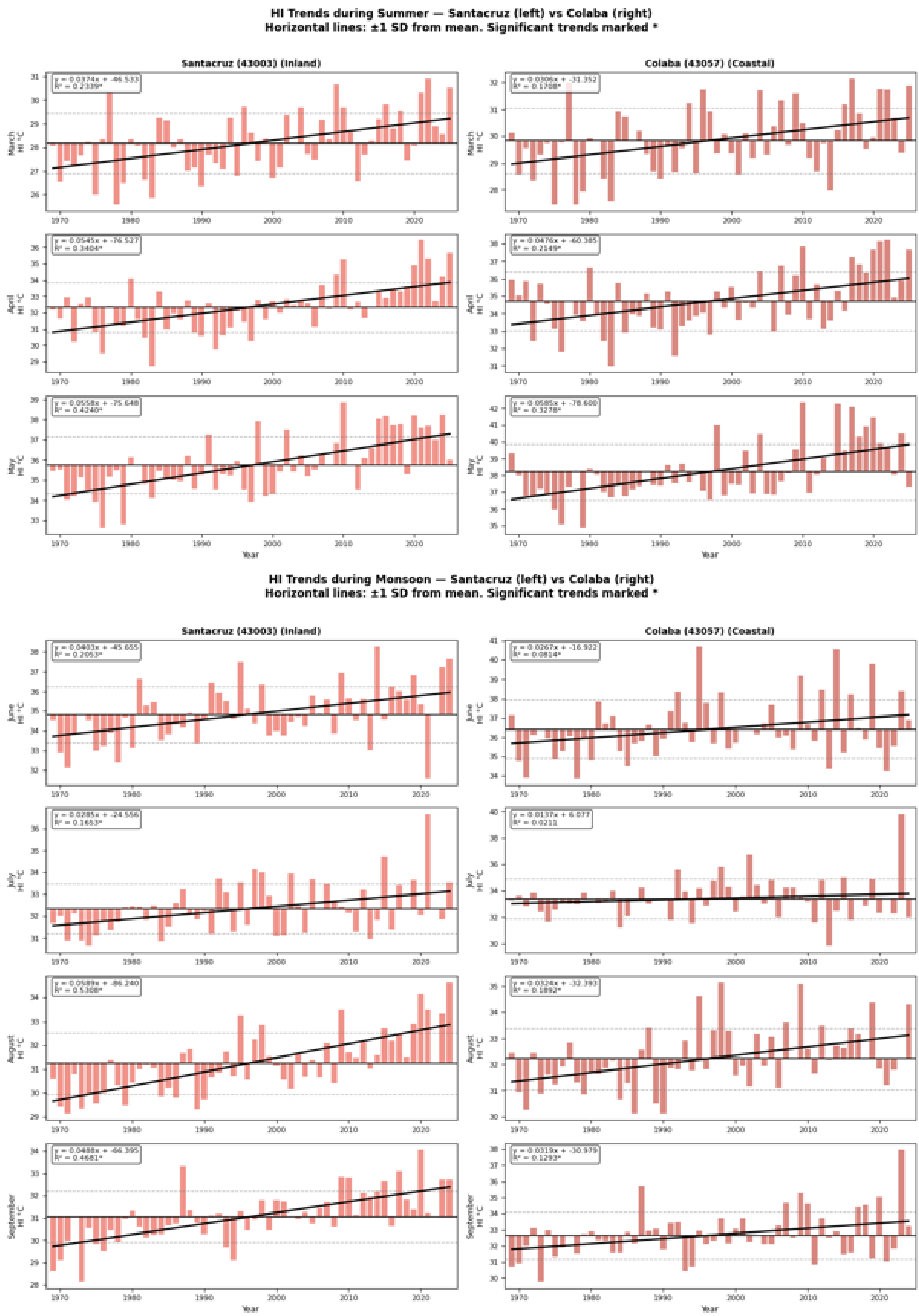

Figure 6.6: Monthly Heat Index during Summer and Monsoon

The graphs further highlight a long-term increasing trend in HI values, as indicated by the positive slope in the regression lines across most months. This suggests that heat stress conditions are gradually worsening over time, likely due to climate change and urbanization effects such as the Urban Heat Island (UHI) phenomenon. Coastal areas like Colaba consistently exhibit higher HI values compared to inland regions, primarily due to higher relative humidity influenced by proximity to the Arabian Sea.

Additionally, the variability in HI values, represented by fluctuations around the mean and standard deviation lines, indicates the occurrence of extreme heat events, especially in recent years. These extremes pose serious risks to human health, productivity, and urban livability, particularly for vulnerable populations. Overall, the analysis emphasizes that the combined effect of temperature and humidity during summer and monsoon seasons significantly amplifies heat stress, making these periods critical for heat mitigation and adaptation strategies in cities like Mumbai.

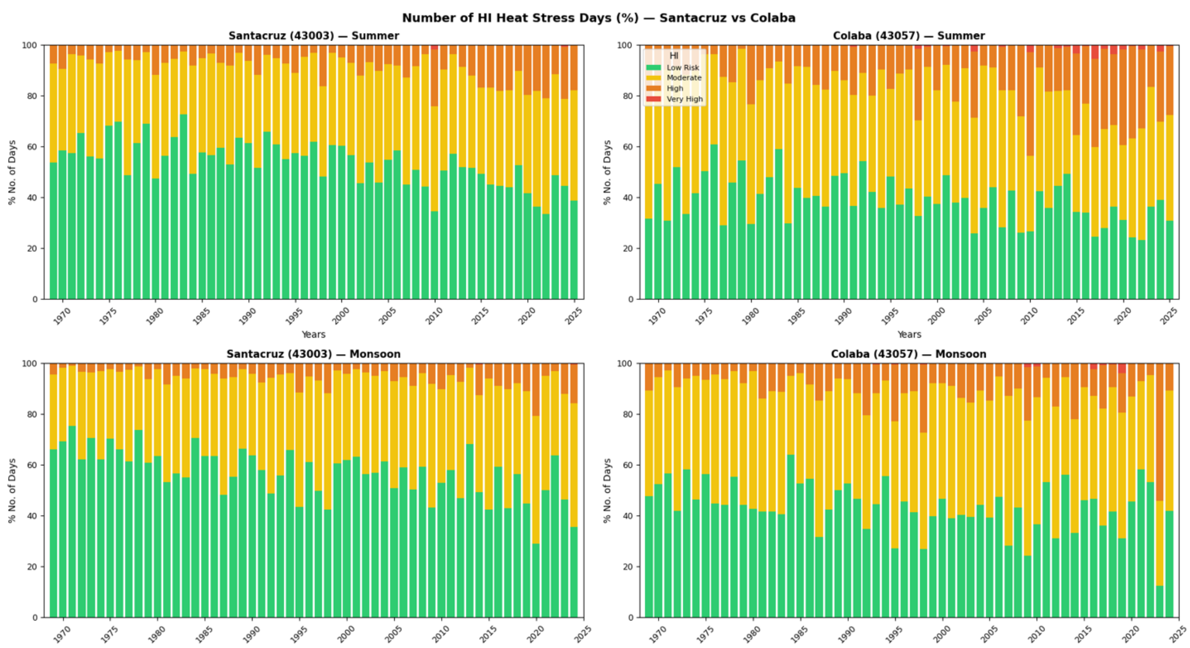

Figure 6.7: Annual heat index trend

The annual trend analysis in Figure 6.7 indicates a gradual increase in heat stress levels over time. This suggests the influence of long-term climatic changes and urbanization, contributing to rising thermal discomfort.

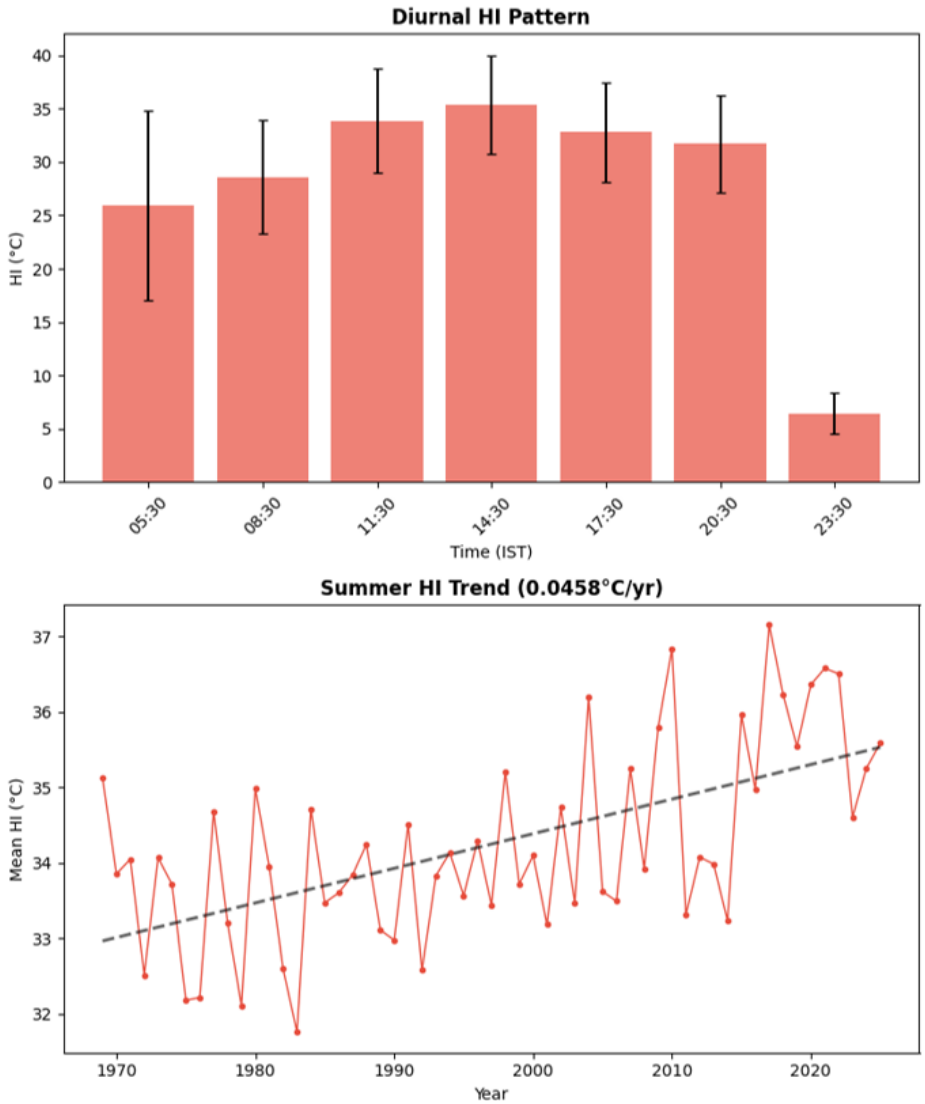

Figure 6.8: Diurnal + temporal trends

The diurnal variation of the Heat Index in Figure 6.8 indicates that heat stress reaches its peak during the afternoon hours, primarily due to maximum solar radiation and elevated air temperatures. During this period, the combined effect of intense sunlight and surface heating leads to a significant increase in thermal discomfort and physiological strain.

Even during the nighttime hours, the Heat Index remains relatively high, especially in urban areas. This is mainly due to high humidity levels, which limit the body's ability to cool through evaporation (sweating). Additionally, the presence of urban heat island effects, such as heat retention by buildings and paved surfaces, further reduces nighttime cooling. As a result, the lack of adequate nocturnal relief prolongs heat exposure, increasing the risk of heat-related stress and health impacts.

6.2.2 Feature Analysis

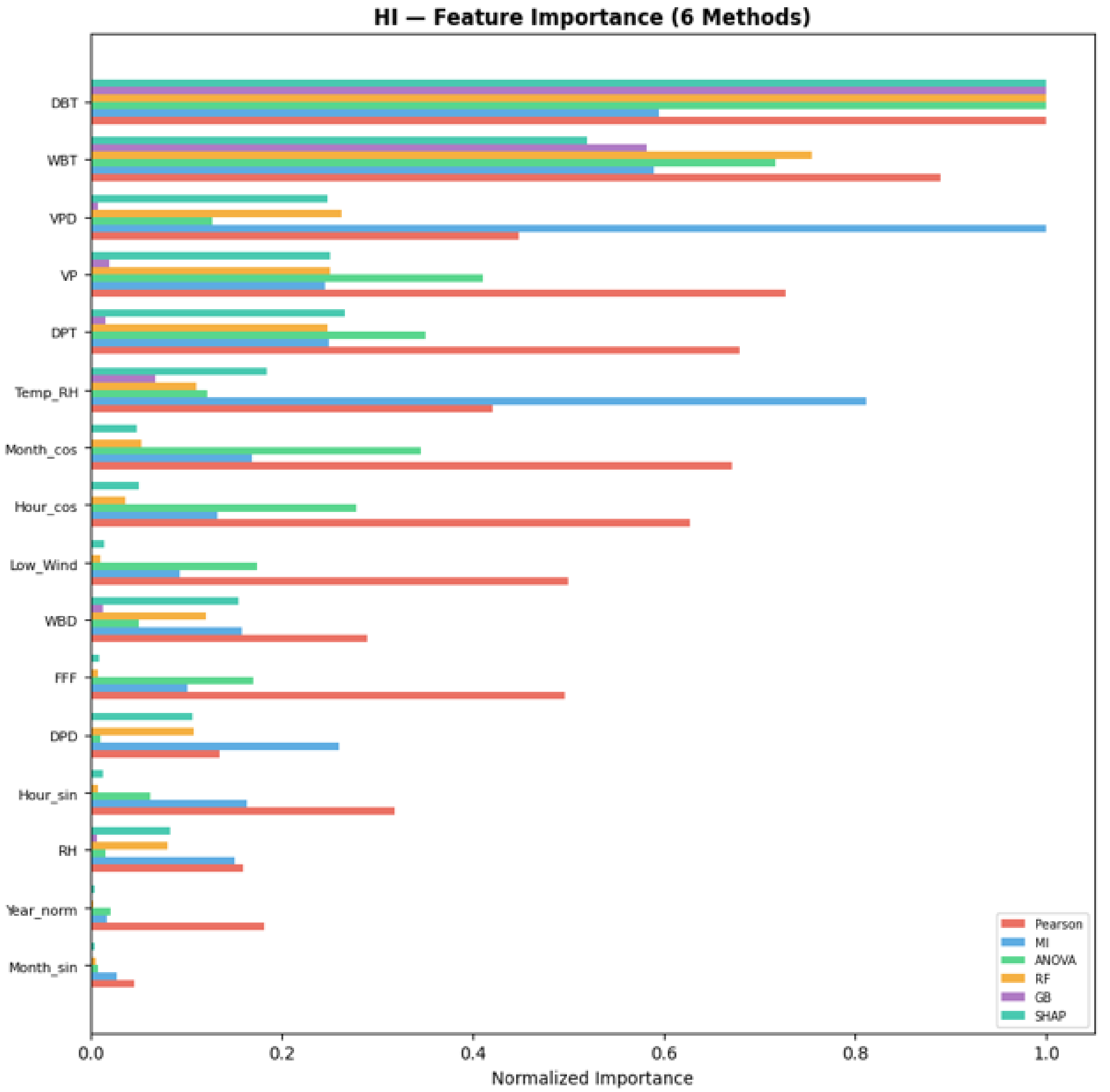

Figure 6.9: Feature importance comparison – 6 methods

Feature importance analysis in Figure 6.9 indicates that temperature and humidity are the most significant contributors to heat stress classification. Wind speed and pressure have comparatively lower influence. This confirms the dominant role of thermal and moisture variables in determining human heat stress.

6.2.3 Model Results/Evaluation

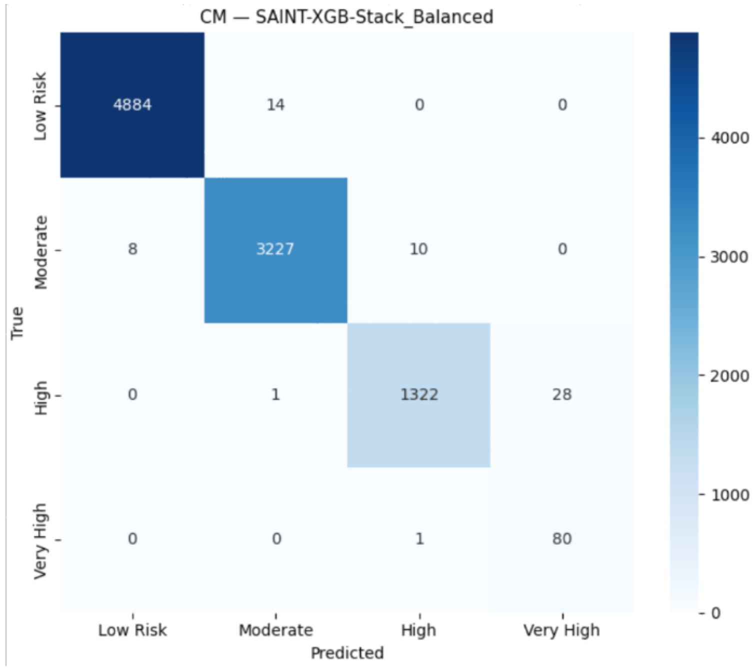

Figure 6.10: Confusion matrix for heat stress classification using the proposed SAINT-XGB-Stack model

The confusion matrix illustrates the classification performance of the model across different heat stress categories. The model achieves high accuracy in identifying all classes, with particularly strong performance in detecting high and very high heat stress conditions.

Table 6.3: Test Set Results (Best per model) — Colaba (43057) — HI

Model	Type	Accuracy	F1	Precision	Recall	Kappa	ROC-AUC	Time (s)
SAINT-XGB-Stack	Balanced	0.9935	0.9561	0.9316	0.9894	0.9893	0.9999	4088.6
XGBoost	Balanced	0.9934	0.9525	0.9267	0.9888	0.9891	0.9999	7.9
SAINT	Balanced	0.9708	0.9439	0.9226	0.9734	0.9514	0.9986	1083.1
SVC	Balanced	0.9807	0.9280	0.9058	0.9569	0.9681	0.9992	105.5
KNN	Balanced	0.9316	0.8480	0.8476	0.8484	0.8866	0.9663	0.3
BBAG	Imbalanced	0.9431	0.8190	0.7829	0.9447	0.9071	0.9950	0.9
AdaBoost	Balanced	0.8728	0.8048	0.7707	0.8587	0.7964	0.9419	9.0
LDA	Balanced	0.8334	0.7716	0.7297	0.8911	0.7365	0.9679	0.1
NaiveBayes	Imbalanced	0.7731	0.7367	0.7347	0.7511	0.6338	0.9250	0.0
EasyEnsemble	Imbalanced	0.6199	0.5954	0.6368	0.7796	0.4333	0.8360	3.1
RUSBoost	Imbalanced	0.6543	0.5039	0.5087	0.5506	0.4739	0.8910	0.1

The performance comparison of various machine learning models on the IMD dataset is presented in Table 6.3 . The proposed SAINT-XGB-Stack model achieves the highest performance among all models, with an accuracy of 99.35% and F1-score of 0.9561. This indicates its superior capability in correctly classifying different heat stress levels.

The high ROC-AUC value (0.9999) further demonstrates the model's strong discriminative ability, while the high recall (0.9894) indicates its effectiveness in identifying heat stress conditions with minimal false negatives. Compared to individual models such as XGBoost and SAINT, the stacked ensemble benefits from combining multiple learning approaches, resulting in improved robustness and generalization.

Overall, the results highlight that ensemble learning techniques, particularly the proposed stacking approach, are highly effective for heat stress classification using IMD meteorological data.

Table 6.4: Test Set Results (Best per model) — Santacruz (43003) — HI

Model	**Type**	**Accuracy**	**F1**	**Precision**	**Recall**	**Kappa**	**ROC-AUC**	**Time (s)**
SAINT-XGB-Stack	Balanced	0.9934	0.9470	0.9237	0.9781	0.9877	0.9998	3135.5
XGBoost	Balanced	0.9961	0.9251	0.9210	0.9294	0.9927	0.9999	30.1
SVC	Imbalanced	0.9703	0.8632	0.8429	0.8870	0.9447	0.9993	247.1
KNN	Balanced	0.9381	0.7473	0.7273	0.8862	0.8839	0.9697	1.5
BBAG	Imbalanced	0.9008	0.7234	0.7028	0.8592	0.8149	0.9830	1.2
AdaBoost	Balanced	0.8562	0.7207	0.6931	0.8020	0.7416	0.8563	30.8
SAINT	Balanced	0.8735	0.6734	0.6929	0.8744	0.7557	0.9231	679.7
LDA	Balanced	0.7795	0.6492	0.6114	0.8236	0.6269	0.9591	0.3
NaiveBayes	Imbalanced	0.7722	0.6168	0.6043	0.6577	0.5890	0.8987	0.0
RUSBoost	Imbalanced	0.5349	0.4424	0.5814	0.5118	0.2875	0.7756	0.1
EasyEnsemble	Imbalanced	0.4154	0.3799	0.4967	0.5842	0.1608	0.7913	3.9

Table 6.4 presents the performance comparison of various machine learning models on the IMD Santacruz dataset. The SAINT-XGB-Stack model achieves the highest overall performance, with an accuracy of 99.34% and F1-score of 0.9470, demonstrating its strong capability in classifying heat stress levels.

Although XGBoost attains slightly higher accuracy (99.61%), the proposed stacked model provides a better balance between precision and recall, resulting in improved F1-score and more reliable classification performance. The high ROC-AUC values across top models indicate excellent discrimination ability between different heat stress categories.

The results further highlight that ensemble-based approaches outperform individual models by effectively capturing complex relationships within meteorological data. Overall, the proposed model demonstrates robust and consistent performance, making it suitable for real-world heat stress monitoring applications.

The consistent performance across both coastal (Colaba) and inland (Santacruz) stations confirms the generalizability of the proposed model.

Chapter 7

Conclusion

This project presents a comprehensive framework for analyzing and predicting heat stress conditions in Mumbai using both ERA5 reanalysis data and IMD observational datasets. The study successfully integrates meteorological data analysis, feature engineering, and advanced machine learning techniques to model complex environmental interactions influencing heat stress.

The exploratory analysis revealed strong seasonal and temporal patterns, with significantly elevated heat stress levels during the summer and monsoon seasons due to the combined effects of temperature and humidity. The inclusion of engineered features such as lag variables, rolling statistics, and cyclical time encoding further enhanced the model's ability to capture temporal dependencies.

Among the various models evaluated, the SAINT-XGB-Stack ensemble model demonstrated superior performance, achieving high accuracy, F1-score, and near-perfect ROC-AUC values across both Colaba and Santacruz stations. The results highlight the effectiveness of ensemble learning in improving prediction robustness and generalization capability. Additionally, the model exhibited minimal overfitting, confirming its reliability for real-world applications.

The forecasting results indicate a high frequency of heat stress days, emphasizing the increasing impact of climate change and urbanization, particularly in densely populated coastal cities like Mumbai. The integration of prediction and classification approaches provides a scalable and efficient system for early warning and risk assessment.

Overall, this project bridges the gap between traditional climate analysis and modern data-driven methodologies. The proposed framework can serve as a valuable tool for public health planning, urban policy development, and climate resilience strategies. Future work may focus on incorporating real-time data, expanding to multiple cities, and integrating deep learning approaches for further improvement in predictive accuracy.

References

[1] F. Briegel, S. Schrodi, M. Sulzer, T. Brox, J. G. Pinto, and A. Christen, "Deep learning enables city-wide climate projections of street-level heat stress," *Urban Climate*, vol. 62, p. 102564, 2025.

[2] G. Suthar, N. Kaul, S. Khandelwal, and S. Singh, "Predicting land surface temperature and examining its relationship with air pollution and urban parameters in Bengaluru: A machine learning approach," *Urban Climate*, vol. 53, p. 101830, 2024.

[3] G. Suthar, R. P. Singhal, S. Khandelwal, and N. Kaul, "Spatiotemporal variation of air pollutants and their relationship with land surface temperature in Bengaluru, India," *Remote Sensing Applications: Society and Environment*, vol. 32, p. 101011, 2023.

[4] K. Kacker, A. Utpal, S. Singh, P. Srivastava, M. Mukherjee, S. S. Subramanian, and R. Rakesh, "A framework for impact-based heat stress warning system for a coastal city in India," *Scientific Reports*, 2026.

[5] J. Kong, Y. Zhao, J. Carmeliet, and C. Lei, "Urban heat island and its interaction with heatwaves: A review of studies on mesoscale," *Sustainability*, vol. 13, no. 19, p. 10923, 2021.

[6] X. Zhang et al., "Deep learning-based 500 m spatio-temporally continuous air temperature generation by fusing multi-source data," *Remote Sensing*, vol. 14, p. 3536, 2022.

[7] Y. Wu, B. Teufel, L. Sushama, S. Belair, and L. Sun, "Deep learning-based super-resolution climate simulator-emulator framework for urban heat studies," *Geophysical Research Letters*, vol. 48, 2021.

[8] K. Ashwini, B. S. Sil, A. A. Kafy, H. A. Altuwaijri, H. Nath, and Z. A. Rahaman, "Harnessing machine learning algorithms to model the association between land use/land cover change and heatwave dynamics for enhanced environmental management," *Land*, vol. 13, p. 1273, 2024.

[9] D. I. V. Domeisen et al., "Prediction and projection of heatwaves," *Nature Reviews Earth & Environment*, 2022.

[10] N. Alinasab et al., "A measurement-based framework integrating machine learning and morphological dynamics for outdoor thermal regulation," *International Journal of Biometeorology*, vol. 69, pp. 1645–1662, 2025.

[11] F. Shafiq, A. Zafar, M. U. G. Khan, S. Iqbal, A. S. Albesher, and M. N. Asghar, "Extreme heat prediction through deep learning and explainable AI," *PLOS ONE*, vol. 20, no. 3, p. e0316367, 2025.

[12] V. Jacques-Dumas, F. Ragone, P. Borgnat, P. Abry, and F. Bouchet, "Deep learning-based extreme heatwave forecast," *Frontiers in Climate*, vol. 4, p. 789641, 2022.

[13] R. A. Berk, A. Braverman, and A. K. Kuchibhotla, "Algorithmic forecasting of extreme heat waves," *arXiv preprint arXiv:2409.18305*, 2024.

[14] G. Miloshevich, B. Cozian, P. Abry, P. Borgnat, and F. Bouchet, "Probabilistic forecasts of extreme heatwaves using convolutional neural networks in a regime of lack of data," 2023.

[15] F. A. Chishtie et al., "Advancing heatwave forecasting via distribution-informed graph neural networks (DI-GNNs): Integrating extreme value theory with GNNs," 2024.

[16] B. Vinayak, H. S. Lee, and S. Gedam, "Prediction of land use and land cover changes in Mumbai city using remote sensing data and a multilayer perceptron based Markov chain model," *Sustainability*, vol. 13, p. 471, 2021.

[17] G. Suthar, S. Singh, N. Kaul, S. Khandelwal, and R. P. Singhal, "Prediction of maximum air temperature for defining heat wave in Rajasthan and Karnataka states of India using machine learning approach," 2023.

[18] G. Suthar, S. Singh, N. Kaul, and S. Khandelwal, "Diurnal variation of air pollutants and their relationship with land surface temperature in Bengaluru and Hyderabad cities of India," *Remote Sensing Applications: Society and Environment*, vol. 35, p. 101204, 2024.

[19] D. H. García and H. Rezapouraghdam, "Climate change, heat stress and the analysis of its space-time variability in European metropolises," *Journal of Cleaner Production*, vol. 425, p. 138892, 2023.

[20] B. Vinayak et al., "Impacts of future urbanization on urban microclimate and thermal comfort over the Mumbai metropolitan region, India," *Sustainable Cities and Society*, vol. 79, p. 103703, 2022.

[21] G. Suthar et al., "Understanding the multifaceted influence of urbanization, spectral indices, and air pollutants on land surface temperature variability in Hyderabad, India," *Journal of Cleaner Production*, vol. 470, p. 143284, 2024.

[22] B. Nowrouzi-Kia et al., "The impacts of climate change on occupational health and work among outdoor workers: A scoping review," *PLOS Global Public Health*, vol. 6, no. 2, p. e0005888, 2026.

[23] G. Gokmen et al., "Evaluation of student performance in laboratory applications using fuzzy logic," Procedia - Social and Behavioral Sciences, vol. 2, no. 2, pp. 902–909, 2010.

Paper Publications

The following research papers have been published / presented by the authors in connection with the work carried out as part of this project:

Sr.	**Title of Paper**	**Conference / Journal Name**	**Authors**	**Year**
1	Optimizing Brain Tumor MRI Classification Through Transfer Learning and Attention-Boosted CNN Architectures	2nd International Conference on Intelligent Systems and Computational Networks (ICISCN-2026)	Jaya Jeswani et al.	2026
2	A Hybrid SVR, XGBoost and LSTM Model for Temperature Forecasting in Telangana using ERA5 Reanalysis Data	International Conference on Advancing Technology in Engineering and Science (ICATES 2025), IET Conference Proceedings	Jaya Jeswani et al.	2025

Table 7.1: List of Paper Publications

Appendix A: Review Paper

Heat Stress Prediction for Mumbai

Jaya Jeswani
Department of Information Technology
Xavier Institute of Engineering
Mumbai, Maharashtra, India
jayajeswani21@gmail.com

Shravani Jadhav
Department of Information Technology
Xavier Institute of Engineering
Mumbai, Maharashtra, India
shravani24082004@gmail.com

Licia Almeida
Department of Information Technology
Xavier Institute of Engineering
Mumbai, Maharashtra, India
liciaalmeida15@gmail.com

Priyadarshini Sandilyan
Department of Information Technology
Xavier Institute of Engineering
Mumbai, Maharashtra, India
priyasandy2004@gmail.com

Janaki Bal
Department of Information Technology
Xavier Institute of Engineering
Mumbai, Maharashtra, India
janakibal1607@gmail.com

Abstract—Accurate assessment and prediction of heat stress are essential for urban health planning in cities like Mumbai. This study presents a dual-framework approach combining IMD station-based analysis and ERA5-based machine learning modeling. The IMD component uses long-term meteorological data to compute Heat Index (HI) and Physiological Equivalent Temperature (PET), focusing on data preprocessing and imputation. The ERA5 component predicts Wet-Bulb Globe Temperature (WBGT) using engineered features and multiple machine learning and deep learning models within a strict temporal framework. The study integrates feature engineering, model optimization, and uncertainty estimation, with evaluation based on standard regression and classification metrics. Overall, the work highlights the importance of combining observational analysis with advanced predictive modeling to develop reliable heat stress assessment systems.

Keywords—Heat stress prediction, Wet-Bulb Globe Temperature (WBGT), Heat Index (HI), ERA5 reanalysis, IMD data, Machine learning, Deep learning, Time series forecasting, Conformal prediction, PCMCI causal analysis

I. INTRODUCTION

Heat stress is an increasing concern in tropical cities like Mumbai, where high temperature and humidity combine to create dangerous conditions for human health. Traditional temperature-based measures are not sufficient to capture this impact, making advanced indices such as Heat Index (HI), Physiological Equivalent Temperature (PET), and Wet-Bulb Globe Temperature (WBGT) essential for accurate assessment.

This study adopts a dual-framework approach combining IMD station-based analysis and ERA5 reanalysis modeling. The IMD component focuses on long-term observational data from Santacruz and Colaba, emphasizing data preprocessing steps such as imputation, outlier handling, feature selection, and class balancing to analyze historical heat stress patterns. In parallel, the ERA5 component develops a predictive framework for WBGT using extensive feature engineering, temporal validation, and multiple machine learning and deep learning models.

The key contribution of this work lies in integrating observational analysis with advanced predictive modeling under a consistent experimental setup, enabling a structured understanding of heat stress dynamics and supporting the development of reliable heat stress assessment systems.

II. LITERATURE REVIEW

The increasing frequency and intensity of heat waves due to climate change have made heat stress assessment a critical area of research [?]. Traditional approaches, such as those used by meteorological agencies, rely on station-based observations and temperature thresholds, which vary regionally and do not fully capture the combined effects of humidity, radiation, and wind on human heat stress [?]. Studies have shown that urbanization and climate change significantly amplify heat stress, with some regions projected to exceed human adaptability limits under extreme heat and humidity conditions [?, ?, ?].

To overcome these limitations, researchers have adopted composite heat indices and data-driven methods. Machine learning techniques have been used to enhance heat stress indicators and capture complex, non-linear relationships between meteorological variables [?]. Recent studies demonstrate the effectiveness of ML and hybrid models in environmental prediction, including high-resolution heat stress projections and real-time risk assessment using ensemble approaches and explainable AI methods such as SHAP [?, ?, ?]. Additionally, ML-based seasonal heatwave forecasting has shown promise in improving early warning systems [?].

However, most existing work focuses either on observational analysis or predictive modeling separately. There remains a need for integrated frameworks that combine long-term observational data with advanced machine learning approaches for more reliable heat stress assessment and prediction.

III. METHODOLOGY

This section presents the complete pipeline for heat stress analysis and prediction, beginning with data collection and preprocessing, followed by feature engineering and heat-stress index computation, and concluding with model-ready dataset preparation. The IMD-based framework focuses on long-term observational analysis using derived indices such as HI and PET, while the ERA5-based framework supports WBGT-driven predictive modeling. Together, these stages form an integrated system that enhances the reliability and effectiveness of heat stress assessment in Mumbai.

A. Overall Framework

This study adopts a dual-framework approach to analyze and predict heat stress conditions in Mumbai by integrating both observational and data-driven methodologies. The first component utilizes long-term station-based data from the India Meteorological Department (IMD), collected from the Santacruz and Colaba stations, to perform a detailed historical analysis of heat stress using derived indices such as Heat Index (HI) and Physiological Equivalent Temperature (PET). This branch focuses on understanding temporal trends, extreme events, and class-based heat stress distributions through rigorous data preprocessing and statistical analysis.

The second component leverages ERA5 reanalysis data to build a predictive modeling framework, where Wet-Bulb Globe Temperature (WBGT) is used as the primary indicator of human heat stress. This pipeline incorporates feature engineering, temporal validation, and machine learning techniques to model and forecast heat stress conditions. Together, the IMD-based analytical framework and the ERA5-based predictive framework provide a comprehensive and scalable system that combines historical insight with advanced predictive capability for urban heat stress assessment.

B. IMD Data Preparation and Heat-Stress Analysis

The IMD component is based on long-term hourly meteorological observations from the Santacruz (43003) and Colaba (43057) stations in Mumbai. These stations were selected to capture the contrast between inland suburban and coastal climatic conditions. Multiple CSV files covering different year ranges are merged to form a continuous historical dataset spanning 1969 to 2025. Duplicate entries arising from overlapping records are removed using station, date, month, year, and hour identifiers. The IMD synoptic hour (HR) codes are then converted into standard IST datetime format, enabling proper temporal alignment with daily weather cycles.

Following data integration, a comprehensive preprocessing pipeline is applied to improve data quality. Missing values are addressed using a combination of statistical and machine learning-based imputation methods, including mean/median filling, KNN, and MICE approaches. Outliers are treated using an IQR-based filtering technique to remove extreme inconsistencies. These steps are essential due to the long temporal span of the dataset, which introduces gaps and irregularities across key variables such as temperature, humidity, wind speed, rainfall, and cloud cover. The cleaned dataset serves as the foundation for subsequent heat-stress computations.

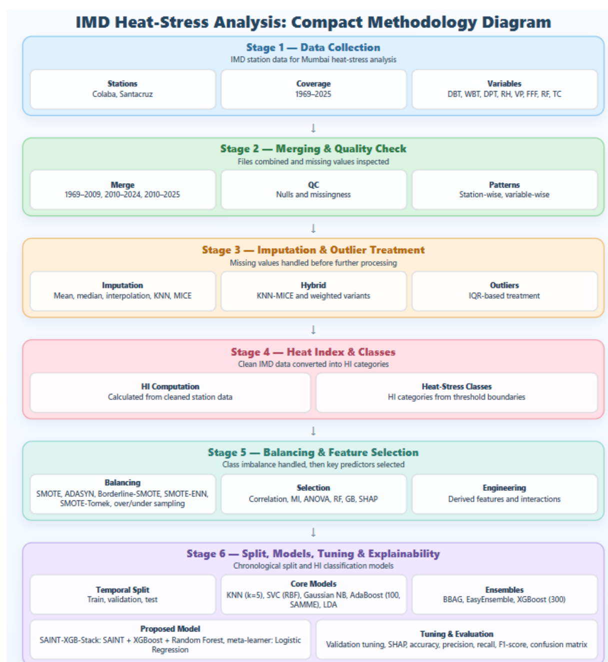

Figure: Overview of the IMD-Based Methodology for Heat Stress Analysis

Using the processed IMD data, two human-centric heat stress indices are computed: Heat Index (HI) and Physiological Equivalent Temperature (PET). HI is calculated using the

NOAA Rothfusz regression based on temperature and relative humidity, while PET incorporates additional environmental factors such as radiation effects, cloud cover, and time of day to provide a more comprehensive thermal comfort measure. Both indices are further transformed into categorical heat-stress classes (e.g., low to extreme risk for HI and comfortable to very hot for PET), enabling classification-based analysis of historical heat conditions.

To address the imbalance in extreme heat categories, various resampling techniques are applied during model preparation. These include SMOTE, ADASYN, Borderline-SMOTE, SMOTE-ENN, SMOTE-Tomek, as well as random over- and under-sampling methods. The dataset is split temporally to preserve real-world conditions, with training data up to 2005, validation from 2006–2015, and testing from 2016–2025. Class balancing is applied only to the training data, while validation and test sets remain in their original imbalanced form to ensure realistic evaluation.

In addition to preprocessing and modeling preparation, the IMD dataset is used for detailed heat-stress analysis across spatial and temporal dimensions. Seasonal and diurnal variations of HI and PET are examined, along with comparative analysis between Santacruz and Colaba stations to understand coastal–inland differences. Long-term trends and class distributions are also analyzed to capture the evolution of extreme heat conditions. Furthermore, feature relevance is assessed using correlation analysis, mutual information, ANOVA, tree-based importance, and SHAP-based interpretability, enabling the identification of key meteorological drivers influencing heat stress.

C. ERA5 Data Preparation and WBGT Computation

The ERA5 component uses daily reanalysis data for Mumbai covering the period from 2015 to 2026. The dataset is first loaded from the prepared ERA5 file, after which the column names are normalized, the date field is identified, and the records are sorted in chronological order.

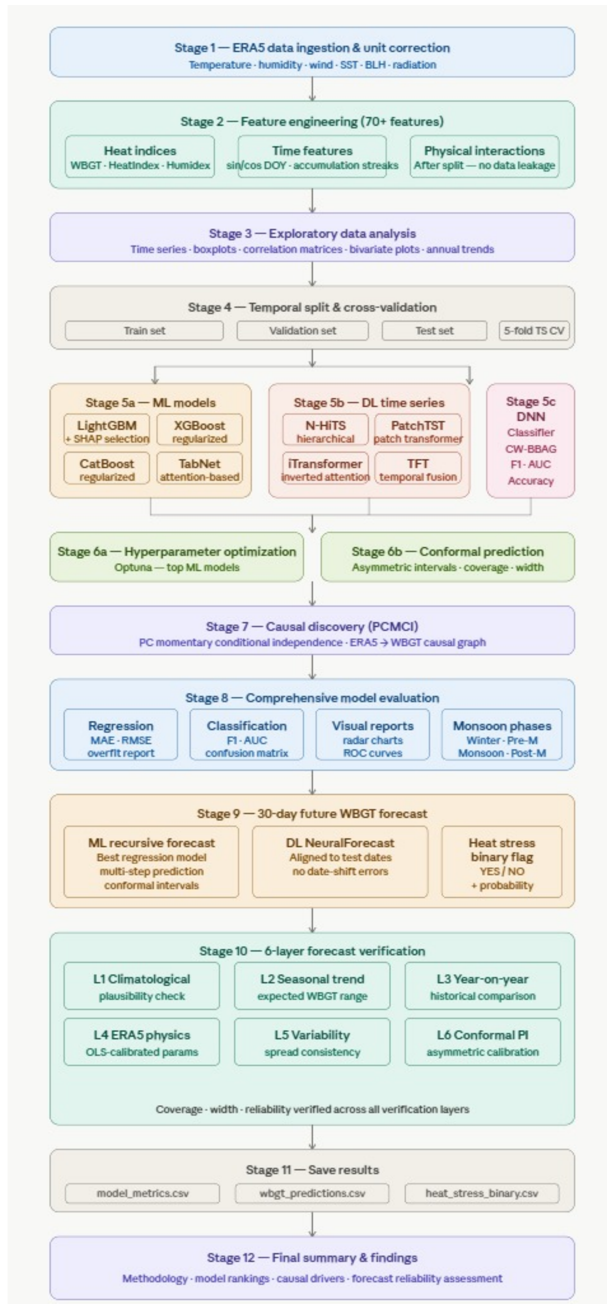

Figure: Overview of the ERA5-Based Methodology for Heat Stress Analysis

After loading, unit correction is applied so that all meteorological variables are expressed in consistent physical units before further processing. For example, geopotential height is converted into meters, pressure variables are converted into hPa, and radiation variables are scaled to W/m^2. The wind components are also combined to derive scalar wind speed and wind direction, while a net-radiation proxy is created to better represent surface energy balance. The main target variable in the ERA5 framework is Wet-Bulb Globe Temperature (WBGT).

In addition to the continuous WBGT value, the ERA5 pipeline also generates derived heat-stress labels for later analysis. These include WBGT-based class categories and a binary heat-stress indicator, which support both regression and classification formulations in the study. This makes the dataset suitable for learning heat-stress conditions as a structured prediction problem rather than relying on temperature alone. The ERA5 workflow is designed around physically meaningful atmospheric variables rather than simple temperature-only predictors.

D. Feature Engineering and Temporal Split

The ERA5 dataset is transformed into a richer modeling table through extensive feature engineering. The final predictor set includes lagged variables, rolling statistics over multiple time windows, anomaly features, physical interaction terms, seasonal encodings, and monsoon-phase indicators. These engineered variables are designed to capture temporal dependence, seasonal variability, and atmosphere–surface interactions that influence heat stress in Mumbai. At the same time, leakage-prone WBGT proxies and direct derived heat indices are excluded from the input feature set so that the model learns only from physically valid predictors.

To preserve the chronological structure of the climate data, the ERA5 records are split strictly by time into training, validation, and test sets. This temporal partitioning avoids look-ahead bias and ensures that model evaluation reflects realistic forecasting conditions rather than random-split performance. The resulting setup supports both model development and unbiased assessment on unseen future periods.

IV. RESULTS AND DISCUSSION

The proposed dual-framework approach was evaluated using both observational analysis from IMD data and predictive modeling using ERA5 reanalysis data. The results demonstrate the effectiveness of combining statistical analysis with machine learning techniques for comprehensive heat stress assessment.

IMD-Based Observational Analysis:

The analysis of long-term IMD data (1969–2025) revealed significant temporal and spatial patterns in heat stress conditions across Mumbai. The computed Heat Index (HI) and Physiological Equivalent Temperature (PET) indicate that heat stress levels frequently exceed safe thresholds, particularly during summer and monsoon seasons.

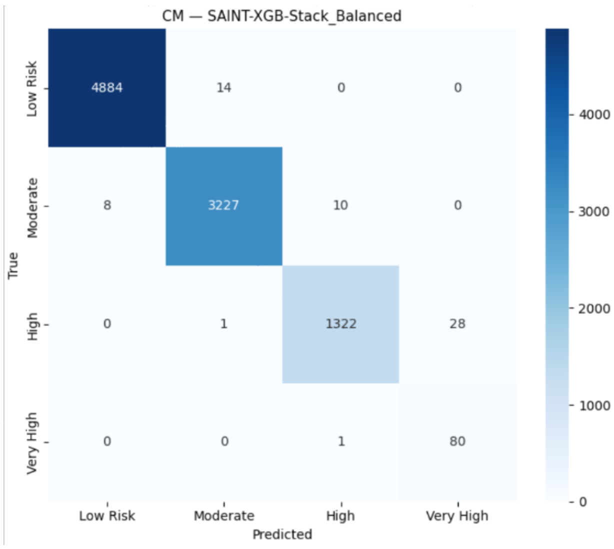

Figure: Confusion matrix for heat stress classification using the proposed SAINT-XGB-Stack model.

The confusion matrix illustrates the classification performance of the model across different heat stress categories. The model achieves high accuracy in identifying all classes, with particularly strong performance in detecting high and very high heat stress conditions.

Seasonal analysis shows that peak heat stress occurs during pre-monsoon and monsoon months due to the combined effect of high temperature and humidity. Diurnal variation analysis indicates that maximum heat stress is observed during afternoon hours, while nighttime conditions remain elevated due to high humidity, reducing thermal comfort.

A comparative analysis between Santacruz and Colaba stations highlights the influence of coastal proximity. Colaba exhibits relatively moderated temperature variations but higher humidity, while Santacruz shows more extreme temperature fluctuations, leading to varying heat stress characteristics.

Long-term trend analysis suggests a gradual increase in extreme heat events, indicating the growing impact of climate change and urbanization. Class-based analysis further confirms the increasing frequency of high-risk heat stress categories over time.

ERA5-Based Predictive Modeling:

The predictive framework using ERA5 data demonstrates strong performance in estimating Wet-Bulb Globe Temperature (WBGT). Machine learning models such as LightGBM, XGBoost, and Random Forest were evaluated using temporal validation.

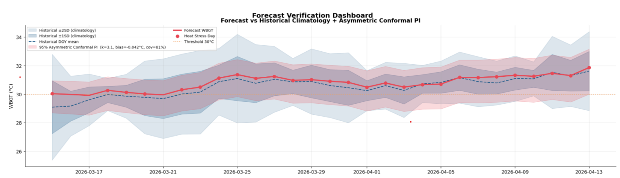

Figure: Forecast verification comparing predicted WBGT with historical climatology and uncertainty intervals

The forecasting results indicate a strong likelihood of persistent heat stress conditions over the next 30 days. The model successfully captures temporal trends and seasonal behavior, while the verification analysis confirms the reliability of predictions. The high number of predicted heat stress days highlights potential risks to human health and emphasizes the importance of proactive measures such as heat warnings, hydration strategies, and urban cooling interventions

The models achieved high accuracy with low prediction error, indicating their effectiveness in capturing complex non-linear relationships between meteorological variables. Evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) remained low, demonstrating reliable predictive capability.

Feature importance analysis revealed that temperature, humidity, solar radiation, and wind-related variables are the most significant contributors to WBGT prediction. The inclusion of lag features and rolling statistics further improved model performance by capturing temporal dependencies.

Residual analysis shows that prediction errors are centered near zero with slight skewness, indicating minimal bias and good generalization capability of the models.

Integrated Insights:

The integration of IMD-based observational analysis with ERA5-based predictive modeling provides a comprehensive understanding of heat stress dynamics. While the IMD framework captures long-term trends and station-level variations, the ERA5 framework enables accurate forecasting of future heat stress conditions.

The results highlight that relying solely on temperature is insufficient, and composite indices such as HI, PET, and WBGT provide a more realistic representation of human thermal stress.

Overall, the proposed system demonstrates robustness, scalability, and practical applicability for urban heat stress assessment and early warning systems.

V. CONCLUSION AND FUTURE WORK

This study presents a dual-framework approach for heat stress analysis in Mumbai by combining long-term IMD observations with ERA5-based predictive modeling. The IMD framework enables historical analysis using HI and PET indices, while the ERA5 pipeline supports WBGT-based prediction using physically meaningful meteorological variables. Both components are developed within a structured pipeline to ensure reliable preprocessing, feature engineering, and evaluation.

The results highlight that integrating observational analysis with data-driven modeling provides a more comprehensive understanding of heat stress dynamics. While the IMD framework captures long-term trends and station-level variations, the ERA5 framework models the multivariate atmospheric conditions influencing heat stress, making the overall system robust and scalable.

Future work will focus on improving interpretability using explainable techniques, extending the framework to other regions, and incorporating additional environmental variables. Further integration with real-time data and advanced modeling approaches can enhance its applicability for heat stress forecasting and climate adaptation planning.

V. REFERENCES

[1] India Meteorological Department, "Heat wave conditions over India: Climatology and trends," *Mausam*, 2020.

[2] G. A. Meehl and C. Tebaldi, "More intense, more frequent, and longer lasting heat waves in the 21st century," *Science*, vol. 305, no. 5686, pp. 994–997, 2004.

[3] S. C. Sheridan and C. C. Lee, "Extending the utility of heat stress indicators using a machine learning approach," *Applied Geography*, vol. 93, pp. 98–106, 2018.

[4] "Global projections of heat stress at high temporal resolution using machine learning," *Earth System Science Data*, 2025.

[5] G. Katavoutas and D. Founda, "Response of urban heat stress to climate change and urbanization in Athens," *Atmosphere*, 2019.

[6] J. S. Pal and E. A. B. Eltahir, "Future temperature in southwest Asia projected to exceed a threshold," *Nature Climate Change*, 2016.

[7] C. Raymond *et al.*, "The emergence of heat and humidity too severe for human tolerance," *Science Advances*, 2020.

[8] "Measurement-based framework for outdoor thermal regulation," *International Journal of Biometeorology*, 2025.

[9] "Machine learning framework for environmental health risk mapping," *Scientific Reports*, 2025.

[10] "Seasonal heatwave forecasting using machine learning," *Stochastic Environmental Research and Risk Assessment*, 2021.